Course Notes

Math 115B

2 July 2026

Introduction

These are typesetted notes for Math 115B taught in Spring 2026 by Professor Ziyang Gao, written by me. First of all, I would like give a huge thanks to Vicky for consistently providing her notes, and also Kenneth, Elia, and Hyunjae for providing notes whenever Vicky could not make it to class.

This course directly succeeds Math 115AH taught by Professor Will Conley in Winter 2026, which has an excellent set of notes typed up by Professor Conley, and with proofs written in by Vicky, which I will assume and refer to throughout these notes. Hopefully these will come in one package with these notes.

These notes roughly follow the lectures given by the professor, and the course follows the book by Friedberg, Insel, Spence. However, I like to add in remarks which either I think build intuition, connect to other areas of math, give generalizations, or are otherwise interesting tidbits of math. In particular, while I recognize it is hard to say anything meaningful about infinite dimensional spaces without a topology, I still think it lays important groundwork for future courses to know which facts only hold in finite dimensions, and roughly why.

For certain topics, I have taken the liberty of altering the presentation. The sections on quadratic forms (Lectures 19-21) deviate greatly from the lectures as I decided to rewrite everything, more closely following Axler's and Treil's books. For some sections on Jordan form (Lectures 22-23) I have also chosen to follow those books more closely.

Contents

Lecture 1 March 30

Omitted.

Lecture 2 April 1, Dual Spaces

Review

Recall the following facts about linear transformations and matrices:

Proposition 2.1

Let V, W be vector spaces with $\dim(V) = n$, $\dim(W) = m$, so $V \simeq F^n$ and $W \simeq F^m$. Let $\mathcal{B} = \{v_1, \ldots, v_n\}$ and $\mathcal{C} = \{w_1, \ldots, w_m\}$ be bases for V and W, respectively.

Recall $\mathcal{L}(V, W)$ is the space of linear transformations between V and W and $M_{m\times n}(F)$ is the space of $m \times n$ matrices, and both spaces have a vector space structure. Furthermore there exists an isomorphism between them given by the following:

$$T \in \mathcal{L}(V, W) \quad \longleftrightarrow \quad [T]_{\mathcal{B}}^{\mathcal{C}} = \begin{pmatrix} [T(v_1)]_{\mathcal{C}} & \cdots & [T(v_n)]_{\mathcal{C}} \end{pmatrix}$$

Thus we have

$$\dim \mathcal{L}(V, W) = mn = \dim(W) \cdot \dim(V)$$

Proposition 2.2

If $V = W$, then T sometimes has an inverse T^{-1} satisfying $T \circ T^{-1} = T^{-1} \circ T = \mathrm{id}_V$, and similarly for matrices. Then in this case $[T^{-1}]_{\mathcal{B}}^{\mathcal{C}} = ([T]_{\mathcal{B}}^{\mathcal{C}})^{-1}$. When T is invertible we have T is injective and surjective, or $\ker(T) = \{0\}$ and $\mathrm{im}(T) = V$.

Dual Spaces

Let V be a vector space over a field F.

Definition 2.3 (Dual Space) — The **dual space** of V, denoted V^*, is defined by $V^* := \mathcal{L}(V, F)$. The elements of V^* are often called linear functionals.

Proposition 2.4

If $\dim(V) = n$, then $\dim(V^*) = n$.

Proof. This immediately follows from the fact $\dim \mathcal{L}(V, F) = \dim(V) \cdot \dim(F) = n \cdot 1 = n$. $\square$

Example 2.5

Let $V = M_{n\times n}(F)$, and consider a linear functional $f : M_{n\times n}(F) \to F$. There are a few interesting cases to consider:

$f(A) = \mathrm{tr}(A)$, the trace function. This is invariant under change of bases (or similar matrices).

Note that $A \mapsto \det(A)$ is not a linear function since $\det(A + B) \neq \det(A) + \det(B)$.

Remark 2.6. The fact that the trace function is invariant under change of bases means that it is well-defined for abstract linear operators. That is, we can take $T : V \to V$, and if V is finite dimensional, we can choose any basis $\mathcal{B}$ and consider the matrix $A = [T]_{\mathcal{B}}^{\mathcal{B}}$ and let $f(T) = \mathrm{tr}(A)$, which is well-defined since it is independent of the choice of basis.

Example 2.7

Let $V = \{f : [0, 2\pi] \to \mathbb{R}\}$. Define $\hat{f} : V \to \mathbb{R}$ by

$$\hat{f}(f) = \frac{1}{2\pi}\int_0^{2\pi} f(t)\cos(t)\,dt$$

Then $\hat{f}$ is a linear functional.

Remark 2.8. This is closely related to the Fourier transform. Here the vector space is infinite-dimensional, meaning we need functional analysis to extract meaningful information, so we will focus on the finite dimensional case.

Now the question is, given a basis of V, can we construct a basis of V^*?

Claim 2.9 — Given a finite basis $\mathcal{B}$ of V, there exists a canonical choice of basis for V^*, called the dual basis, denoted by $\mathcal{B}^*$.

Theorem 2.10

Let $\mathcal{B} = \{v_1, \ldots, v_n\}$. Define $f_i : V \to F$ for $i = 1, \ldots, n$ as follows:

$$f_i(a_1 v_1 + \cdots + a_n v_n) = a_i \quad \forall\, a_1, \ldots, a_n \in F$$

Equivalently, we can define

$$f_i(v_j) = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

and extend by linearity.

Let $\mathcal{B}^* = \{f_1, \ldots, f_n\}$. Then $\mathcal{B}^*$ is a basis of V^* and in particular for any $f \in V^*$ we have

$$f = \sum_{i=1}^{n} f(v_i) f_i$$

Sometimes we write $v_i^* := f_i \in V^*$

Proof. Since we know $\dim(V^*) = n$ and $\mathcal{B}^*$ has length n, to show it is a basis, it suffices to show it is a spanning set. Let $f \in V^*$. Consider $g := \sum_{i=1}^n f(v_i) f_i \in V^*$. We wish to show $f = g$.

For any $j \in \{1, \ldots, n\}$, we have

$$g(v_j) = \left(\sum_{i=1}^{n} f(v_i) f_i \right)(v_j) = \sum_{i=1}^{n} f(v_i) f_i(v_j) = \sum_{i=1}^{n} f(v_i) \delta_{ij} = f(v_j)$$

Thus f and g agree on $\mathcal{B}$, and by linearity, it follows $f = g$ for all $v \in V$. Thus it follows that $\mathcal{B}^*$ spans V^* and the formula for f holds. $\square$

Example 2.11

Let $V = \mathbb{R}^2$ and $\mathcal{B} = \{e_1, e_2\} = \{(1,0), (0,1)\}$. Then

- $e_1^* : \mathbb{R}^2 \to \mathbb{R}$, and since $(a_1 e_1 + a_2 e_2) \mapsto a_1$ (by the first definition), we have $(a_1, a_2) \mapsto a_1$.
- $e_2^* : \mathbb{R}^2 \to \mathbb{R}$, and since $(a_1 e_1 + a_2 e_2) \mapsto a_2$, we have $(a_1, a_2) \mapsto a_2$.

Now suppose we have a different basis $\mathcal{B} = \{v_1, v_2\}$, where $v_1 = (2,1)$, $v_2 = (3,1)$. To find $\mathcal{B}^*$, we use the second definition:

$$\begin{cases} v_1^*(2,1) = v_1^*(2e_1 + e_2) = 1 \\ v_1^*(3,1) = v_1^*(3e_1 + e_2) = 0 \end{cases}$$

We can solve this system to get

$$\begin{cases} v_1^*(e_1) = v_1^*(1,0) = -1 \\ v_1^*(e_2) = v_1^*(0,1) = 3 \end{cases} \iff v_1^*(a,b) = -a + 3b \iff v_1^* = -e_1^* + 3e_2^*$$

Similarly, for v_2, we can obtain

$$v_2^*(a,b) = a - 2b \iff v_2^* = e_1^* - 2e_2^*$$

Lecture 3 April 3, Dual Maps

Definition 3.1 (Dual Map) — Let V, W be vector spaces, and $T : V \to W$ be linear. Then there exists a canonical map $T^* : W^* \to V^*$, called the **dual map** of T, defined by

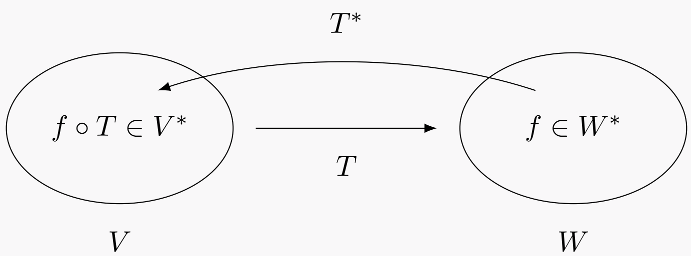
$$T^*(f) = f \circ T \quad f \in W^*$$

Sometimes it is also called the **transpose** of T, and may be denoted T^T or T'.

Remark 3.2. V, W need not be finite-dimensional for this to be defined. Note $f \circ T : V \to F$ is linear so $f \circ T \in V^*$.

We sometimes call T^* the **pullback** of f along T. A common idea in math is that we have two spaces with a nice transformation between them (in this case linear), and we have an object defined on one space, which we would like to "move" to the other space via the map we have.

Here, we have a linear functional f defined on W, and a natural question is to ask is how can we similarly define a linear functional on V through the map T. The natural way to do this is to take $f : W \to F$ and for elements $v \in V$ we feed it through T first and then to f to get $f(T(v))$.



Similar notions exist in measure theory (and probability) where you have a measurable function between measurable spaces, and you have a measure defined on one space and you usually want to do a pushforward to define a measure on the other space. In differential geometry you have smooth maps between manifolds, and the natural objects to pullback are differential forms.

Theorem 3.3

Let V, W be finite dimensional with $\dim(V) = n$ and $\dim(W) = m$. Let $\mathcal{B} = \{v_1, \dots, v_n\}$ and $\mathcal{C} = \{w_1, \dots, w_m\}$ be bases for V and W, respectively. Then,

$$[T^*]_{\mathcal{C}^*}^{\mathcal{B}^*} = \left([T]_{\mathcal{B}}^{\mathcal{C}}\right)^T$$

Proof. We have $[T^*]_{\mathcal{C}^*}^{\mathcal{B}^*} = \begin{pmatrix} [T^*(w_1^*)]_{\mathcal{B}^*} & \cdots & [T^*(w_m^*)]_{\mathcal{B}^*} \end{pmatrix}$. Then fix $w_i^* \in \mathcal{C}^*$. We consider $T^*(w_i^*)$ and expand it out in terms of the $\mathcal{B}^*$ using Theorem 2.10 from last time:

$$T^*(w_i^*) = w_i^* \circ T = \sum_{j=1}^{n} (w_i^* \circ T)(v_j)\, v_j^*$$

That is,

$$[T^*(w_i^*)]_{\mathcal{B}^*} = \begin{pmatrix} w_i^*(T(v_1)) \\ \vdots \\ w_i^*(T(v_n)) \end{pmatrix}$$

so the whole matrix is

$$[T^*]_{\mathcal{C}^*}^{\mathcal{B}^*} = \begin{pmatrix} w_1^*(T(v_1)) & \cdots & w_m^*(T(v_1)) \\ \vdots & \ddots & \vdots \\ w_1^*(T(v_n)) & \cdots & w_m^*(T(v_n)) \end{pmatrix}$$

Now we look at the rows, which are given by

$$\begin{pmatrix} w_1^*(T(v_j)) & \cdots & w_m^*(T(v_j)) \end{pmatrix}$$

But recall each w_i^* is defined as a Kronecker delta - it is equal to 1 for w_i and 0 for any w_j for $j \neq i$, in other words it picks out the ith coordinate of the vector in the $\mathcal{C}$ basis. That is, the jth row is exactly the $\mathcal{C}$-coordinates of $T(v_j)$.

Now recall

$$[T]_{\mathcal{B}}^{\mathcal{C}} = \begin{pmatrix} [T(v_1)]_{\mathcal{C}} & \cdots & [T(v_n)]_{\mathcal{C}} \end{pmatrix}$$

Then it is clear the columns of $[T]_{\mathcal{B}}^{\mathcal{C}}$ correspond to the rows of $[T^*]_{\mathcal{C}^*}^{\mathcal{B}^*}$, or that $[T^*]_{\mathcal{C}^*}^{\mathcal{B}^*} = \left([T]_{\mathcal{B}}^{\mathcal{C}}\right)^T$. □

Corollary 3.4

Let A be $m \times n$ matrix and B an $n \times p$ matrix, so AB is an $m \times p$ matrix. Then $(AB)^T = B^T A^T$.

Proof. Let $T_A : F^n \to F^m$ be defined by $x \mapsto Ax$ and $T_B : F^p \to F^n$ be defined by $y \mapsto By$. Then $T_A \circ T_B$ is defined by $y \mapsto ABy$. Now recall that in the standard basis $\mathcal{S}$ we have: (we should technically distinguish the standard basis for F^m, F^n, F^p but the notation would be too cluttered and it is obvious from context)

$$[T_A]_{\mathcal{S}}^{\mathcal{S}} = A, \quad [T_B]_{\mathcal{S}}^{\mathcal{S}} = B, \quad [T_A \circ T_B]_{\mathcal{S}}^{\mathcal{S}} = AB$$

Then we have

$$[(T_A \circ T_B)^*]_{\mathcal{S}^*}^{\mathcal{S}^*} = (AB)^T$$

But since $(T_A \circ T_B)^* = T_B^* \circ T_A^*$ (exercise), we have

$$[(T_A \circ T_B)^*]_{\mathcal{S}^*}^{\mathcal{S}^*} = [T_B^*]_{\mathcal{S}^*}^{\mathcal{S}^*}[T_A^*]_{\mathcal{S}^*}^{\mathcal{S}^*}$$

and

$$[T_A]_{\mathcal{S}^*}^{\mathcal{S}^*} = A^T, \quad [T_B]_{\mathcal{S}^*}^{\mathcal{S}^*} = B^T$$

It follows that

$$(AB)^T = B^T A^T$$

□

Note that if $V = W$ then both the matrix and its transpose are square. We also have the following property:

Theorem 3.5

Suppose V and W are finite dimensional as before. Then

- T is injective $\iff$ T^* is surjective.
- T is surjective $\iff$ T^* is injective.

Remark 3.6. As expected, there are corresponding statements about the subspaces which characterize injectivity and surjectivity, namely the kernel and image. This is most precisely phrased in terms of the annihilator, which John talked about last quarter (e.g. see Axler Linear Algebra Done Right).

Proof. We prove the first statement.

($\implies$) Assume T is injective. We wish to show T^* is surjective. That is, given $v^* \in V^*$, we want to find $w^* \in W^*$ such that $T^*(w^*) = w^* \circ T = v^*$. Pick a basis $\mathcal{B} = \{v_1, \dots, v_n\}$ and let the dual basis be $\mathcal{B}^* = \{v_1^*, \dots, v_n^*\}$. Then note it suffices to show the result for the basis elements, that is, we want to find $w_i^* \in W^*$ such that $T^*(w_i^*) = v_i^*$ for $i = 1, \dots, n$.

Since $\mathcal{B}$ is linearly independent, and T is injective, $T(\mathcal{B}) = \{T(v_1), \dots, T(v_n)\}$ is also linearly independent. Then extend this to a basis $\mathcal{C}$ for W. Now for a fixed $i \in \{1, \dots, n\}$, define w_i^* by $w_i^*(T(v_i)) = 1$ and $w_i^*(T(v_j)) = 0$ for $i \neq j$. Then we can give w_i^* any values in F for the rest of the basis elements of $\mathcal{C}$, it does not matter. Extending by linearity, we get $w_i^* \in W^*$. Then by definition we have $T^*(w_i^*)(v_j) = w_i^*(T(v_j)) = \delta_{ij}$, so $T^*(w_i^*) = v_i^*$.

Now given $v^* \in V^*$, we can write $v^* = \sum_1^n a_i v_i^*$ for $a_i \in F$. Then take $w^* = \sum_1^n a_i w_i^*$ so $T^*(w^*) = \sum_1^n a_i T^*(w_i^*) = \sum_1^n a_i v_i^* = v^*$, so T^* is surjective, as desired.

($\impliedby$) Assume T^* is surjective. We wish to show T is injective, or $\ker(T) = \{0\}$. Let $v \in V$ such that $T(v) = 0$. Then for any $\eta \in W^*$ we have $T^*(\eta)(v) = \eta(T(v)) = 0$. Now by the surjectivity of T^* for any $\xi \in V^*$ there exists $\eta \in W^*$ such that $T^*(\eta) = \xi$. Thus $\xi(v) = 0$ for all $\xi \in V^*$. Taking a basis for V^*, we have that $v_1^*(v) = \cdots = v_n^*(v) = 0$. It then follows from the definition of the dual basis that $v = 0$. $\square$

Lecture 4 April 6, More Dual Spaces + Direct Sums/Quotients

More Dual Spaces

Let V be a vector space over F. Then note $(V^*)^* = \{f : V^* \to F \mid f \text{ linear}\}$.

Definition 4.1 (Evaluation functional) — Fix $x \in V$. Define $\hat{x} : V^* \to F$ by $\hat{x}(f) = f(x)$. $\hat{x}$ is called the evaluation functional at x because it takes in a linear functional and evaluates it at x. Note that $\hat{x}$ is linear, so $\hat{x} \in V^{**}$.

Proof that $\hat{x}$ is linear. Given $a_1, a_2 \in F$ and $f_1, f_2 \in V^*$ we have $\hat{x}(a_1 f_1 + a_2 f_2) = (a_1 f_1 + a_2 f_2)(x) = a_1 f_1(x) + a_2 f_2(x) = a_1 \hat{x}(f_1) + a_2 \hat{x}(f_2)$. □

But notice we now have a collection maps $\{\hat{x}\}_{x \in V}$ which are indexed by the elements of V. It is natural to ask whether the map which takes an element $x \in V$ to the corresponding map $\hat{x} \in V^{**}$ has any nice properties.

Definition 4.2 — Define $\psi : V \to V^{**}$ by $\psi(x) = \hat{x}$.

Theorem 4.3

If V is finite-dimensional, ψ is a (linear) isomorphism.

Proof. Claim: ψ is linear (This does not require V to be finite-dimensional).

For $a_1, a_2 \in F$ and $x_1, x_2 \in V$ we have $\psi(a_1 x_1 + a_2 x_2) = \widehat{a_1 x_1 + a_2 x_2} \in V^{**}$. Now for any $f \in V^*$ we have

$$(\widehat{a_1 x_1 + a_2 x_2})(f) = f(a_1 x_1 + a_2 x_2) = a_1 f(x_1) + a_2 f(x_2) = a_1 \hat{x}_1(f) + a_2 \hat{x}_2(f)$$

Or that $\widehat{a_1 x_1 + a_2 x_2} = a_1 \hat{x}_1 + a_2 \hat{x}_2$. Finally, we have $\psi(a_1 x_1 + a_2 x_2) = a_1 \psi(x_1) + a_2 \psi(x_2)$.

Claim: ψ is injective

It suffices to show $\ker \psi = \{0\}$. Let $x \in \ker \psi$, so $\psi(x) = \hat{x} = 0$. Then for all $f \in V^*$ we have $\hat{x}(f) = f(x) = 0$. Pick a basis $\mathcal{B} = \{v_1, \ldots, v_n\}$ for V, and let $\mathcal{B}^* = \{v_1^*, \ldots, v_n^*\}$ be the dual basis for V^* (requires finite-dimensional).

Then $v_1^*(x) = \cdots = v_n^*(x) = 0$. Now write $x = a_1 v_1 + \cdots + a_n v_n$, so $v_i^*(x) = a_i = 0$. Thus $x = 0$, and $\ker \psi = \{0\}$.

Finally, since $\dim(V) = \dim(V^*) = \dim((V^*)^*) = n$, we have that ψ must also be surjective and thus bijective, and an isomorphism. □

Remark 4.4. We say there is a **natural** isomorphism between V and V^{**} for finite-dimensional V since the definition of the isomorphism is not dependent on choosing a basis. In contrast, the isomorphism between V and V^* is dependent on choosing some basis and a dual basis.

Note: Professor Gao writes $V^{**} = V$, but they are not equal in the literal sense (one contains linear functionals on a space of linear functionals, and the other one is just vectors). Algebraists just tend to write "=" for isomorphic objects because they have the same properties and behave in the same way, much like how you wouldn't distinguish between the set of natural numbers, and the set containing 1 apple, 2 apples, 3 apples, and so on.

Unfortunately V and V^{**} are usually not isomorphic for infinite-dimensional spaces, but the map ψ is still very interesting to study in functional analysis.

Review of Direct Sums and Quotients

Definition 4.5 (External Direct Sum) — Let V_1, V_2 be vector spaces over F. Then the external direct sum of V_1 and V_2 is given by

$$V_1 \oplus V_2 = \{(v_1, v_2) : v_1 \in V_1,\, v_2 \in V_2\}$$

$V_1 \oplus V_2$ is a vector space with vector addition and scalar multiplication defined component-wise.

Definition 4.6 (Internal Direct Sum) — Let V be a vector space over F. Let W_1, W_2 be subspaces of V. We say V is the internal direct sum of W_1 and W_2 and write $V = W_1 \oplus W_2$ if $V = W_1 + W_2$ and $W_1 \cap W_2 = \{0\}$.

Proposition 4.7

If we consider W_1, W_2 as vector spaces themselves, there is a canonical isomorphism between the external and internal direct sums of W_1 and W_2.

For the proof of this fact and examples of internal and external direct sums, see 115AH HW 3.

Definition 4.8 (Quotient Space) — Let W be a subspace of V. We define the (left) cosets of W to be $v + W := \{v + w : w \in W\}$ for each $v \in V$. We define the quotient space $V/W := \{v + W : v \in V\}$. The quotient space is a vector space with vector addition and scalar multiplication defined by adding/scaling the the v.

These two concepts are central to knowing how to decompose vector spaces. For example, we are often interested in a subspace, and we want to know how to build up the whole space from that subspace.

In fact it is true that the quotient by a subspace is isomorphic to the direct complement of the subspace:

Theorem 4.9

Let W be a subspace of V. Then there exists a subspace W' such that $V = W \oplus W'$. Then V/W is isomorphic to W'.

Proof. See 115AH Worksheet 5. □

Remark 4.10. The professor gives a brief sketch for the finite dimensional case, but as we proved last quarter, this holds for the infinite dimensional case as well. For reasons Conley outlined in the worksheet last quarter, the quotient is a more natural object to work with.

With the isomorphism above, along with the isomorphism between internal and external direct sums, we get

Corollary 4.11

$V \simeq W \oplus V/W$ (as an external direct sum)

Remark 4.12. This is a surprisingly powerful result. It tells us we can always reconstruct an isomorphic copy of the original vector space just by knowing a subspace and the quotient by that subspace. Of course this feels pretty trivial in linear algebra, thanks to how rigid the structure of vector spaces are, allowing us to have concepts of basis and dimension, so any two spaces with the same dimension are isomorphic (although the above result is still true in infinite dimensions).

Unfortunately, you cannot do this for most other algebraic objects like groups or modules - you can construct something from a subspace and a quotient, but it's not guaranteed to be isomorphic.

Lecture 5 April 8, Review of Diagonalization and Eigenstuff

The main motivating question for developing all this theory was if we can find a basis for a linear operator for which its matrix is diagonal.

Definition 5.1 (Eigenstuff and Diagonalizability) — Let V be a vector space over F and $T : V \to V$ be a linear operator. We say a nonzero vector $v \in V$ is an **eigenvector** of T if there exists $\lambda \in F$ such that $T(v) = \lambda v$. λ is called the **eigenvalue** of v.

We call a basis $\mathcal{B}$ for V an **eigenbasis** if each basis vector is an eigenvector of T. Moreover, we say T is **diagonalizable** if there exists an eigenbasis.

We say two matrices $A, B \in M_{n\times n}(F)$ are **similar** if there exists an invertible matrix $Q \in M_{n\times n}(F)$ such that $B = Q^{-1}AQ$. We say A is **diagonalizable** if it is similar to a diagonal matrix

Remark 5.2. You may notice I did not define diagonalizability for an operator as having a basis for which its matrix is diagonal. However, this an equivalent statement. In math, if you have equivalent statements, it can be pretty common to take either as a definition, and then prove the other one from it. I think this one is more generally useful for proving stuff basis-free (and is also a valid definition for infinite dimensions!).

Because of this equivalence, we also have the definition of diagonalizability of matrices is compatible with the diagonalizability of linear operators. Note that this is because any invertible matrix can be thought of as a change of basis matrix, since the columns act as a new basis.

We first prove a fact that justifies our next definition:

Proposition 5.3

If A and B are similar $n \times n$ matrices, then $\det(A - tI_n) = \det(B - tI_n)$

Proof. Let Q be an invertible $n \times n$ matrix such that $B = Q^{-1}AQ$. Then

$$\begin{aligned}\det(B - tI_n) &= \det(Q^{-1}AQ - Q^{-1}tI_nQ)\\ &= \det(Q^{-1}(A - tI_n)Q)\\ &= \det(Q^{-1})\det(A - tI_n)\det(Q)\\ &= \det(A - tI_n)\end{aligned}$$

□

Definition 5.4 (Characteristic polynomial) — Given a matrix $A \in M_{n\times n}(F)$, the **characteristic polynomial** of A is defined by $f_A(t) := \det(A - tI_n)$.

Given a linear operator $T : V \to V$ on a finite dimensional space V, its characteristic polynomial is defined by $f_T(t) = \det([T]_{\mathcal{B}} - tI_n)$ where $n = \dim(V)$ and $\mathcal{B}$ is some basis of V.

Remark 5.5. Because we showed the quantity $\det(A - I_n)$ is invariant under similar matrices (or change of bases), it means the characteristic polynomial is well-defined for general (finite dimensional) linear operators.

Recall the following fact from last quarter:

Proposition 5.6

Let T be as above. λ is a root of $f_T(t) \iff \lambda$ is an eigenvalue of T.

Example 5.7

Consider $T : \mathbb{R}^2 \to \mathbb{R}^2$ defined by $x \mapsto Ax$ for the matrix

$$A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

for some fixed value of θ such that $\sin\theta \neq 0$. Then $f_T(t) = t^2 - (2\cos\theta)t + 1$, which has roots $\cos\theta \pm i\sin\theta = e^{\pm i\theta}$. We see that since $\mathbb{R}^2$ is a vector space over $\mathbb{R}$, since none of the roots are real (since $\sin\theta \neq 0$), there are no eigenvalues, and hence no eigenvectors.

But $\mathbb{R} \subseteq \mathbb{C}$ so if we embed $\mathbb{R}^2 \subseteq \mathbb{C}^2$ as consider as a complex vector space, and consider $T : \mathbb{C}^2 \to \mathbb{C}^2$ it has two eigenvalues $e^{\pm i\theta}$.

Remark 5.8. This is why it is important to know what field you are diagonalizing over, since diagonalizability has a lot to do with the algebraic properties of the field. In particular $\mathbb{C}$, is an algebraically closed field, so every polynomial in $\mathbb{C}$ splits (all roots are in $\mathbb{C}$), which is why linear transformations over $\mathbb{C}$ have much nicer properties.

Definition 5.9 — A polynomial p **splits** over a field F if it can be written as a product of linear factors $p(x) = (x - r_1)\cdots(x - r_n)$ for $r_i \in F$.

Proposition 5.10

If T is a diagonalizable linear operator on a finite dimensional space, then the characteristic polynomial f_T splits.

Definition 5.11 (Eigenspace) — Let T be a linear operator on V. If λ is an eigenvalue of T, then we define the **eigenspace** of T corresponding to λ by $E_\lambda(T) := \ker(T - \lambda I_n)$.

Proposition 5.12

Eigenvectors corresponding to distinct eigenvalues are linearly independent.

Corollary 5.13

If $\lambda_1, \dots, \lambda_k$ are distinct eigenvalues, and $\mathcal{B}_1, \dots, \mathcal{B}_k$ are bases for $E_{\lambda_1}(T), \dots, E_{\lambda_k}(T)$, respectively, then $\mathcal{B}_1 \cup \dots \cup \mathcal{B}_k$ is linearly independent.

Definition 5.14 (Multiplicity of eigenvalues) — Let T be a linear operator on a finite dimensional vector space, and let λ be an eigenvalue of T. We define the **algebraic multiplicity** of λ to be the largest integer k such that $(t-\lambda)^k \mid f_T(t)$.

We define the **geometric multiplicity** of λ to be $\dim E_\lambda(T)$.

Example 5.15

Consider $T : \mathbb{R}^2 \to \mathbb{R}^2$ defined by $x \mapsto Ax$ for

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

Then $f_T(t) = (1-t)^2$, so the eigenvalue $\lambda = 1$ has algebraic multiplicity 2. But $\dim(E_1(T)) = \dim\ker \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = 1$.

Thus we see the two values are not always equal.

Theorem 5.16

Let T be a linear operator on a finite dimensional space.

1. Assume f_T splits over F. Then
$$\sum_{\lambda \text{ eigenvalues}} \text{alg mult}(\lambda) = \deg f_T$$
2. $1 \leq \text{geo mult}(\lambda) \leq \text{alg mult}(\lambda)$
3. T is diagonalizable $\iff$ f_T splits over F and geo mult(λ) = alg mult(λ) for each eigenvalue λ.

Lecture 6 April 10, Invariant Subspaces

Let V be a vector space over F and $T : V \to V$ be a linear operator.

Definition 6.1 (Invariant subspace) — A subspace W of V is called T**-invariant** if $T(W) \subseteq W$.

Remark 6.2. An invariant subspace let's us restrict a linear operator to that subspace: $T|_W : W \to W$.

Example 6.3

A couple of examples of invariant subspaces:

- $\{0\}$
- V
- $\ker(T)$ since $T(\ker(T)) = \{0\} \subseteq \ker(T)$.
- $\mathrm{im}(T)$ since $T(\mathrm{im}(T)) \subseteq T(V) = \mathrm{im}(T)$.

Example 6.4

Eigenspaces are also examples of invariant spaces. Let λ be an eigenvalue of T, and $E_\lambda(T)$ denote the eigenspace corresponding to λ. Then for $v \in E_\lambda(T)$ we have $T(v) = \lambda v \in E_\lambda(T)$ by the definition of subspace.

(Note: this assumes we have already proved $E_\lambda(T)$ is a subspace. This follows from $T(ax+y) = aT(x) + T(y) = a\lambda x + \lambda y = \lambda(ax+y)$ when $x, y \in E_\lambda(T)$.)

If T is diagonalizable, and V is finite dimensional, we can also find a basis $\mathcal{B}$ for which

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}$$

where eigenvalues may be repeated.

Remark 6.5. Note that as we saw last time, eigenspaces need not be 1-dimensional, but each eigenvector in an eigenspace is mapped on to its own span. That is, since $T(v) = \lambda v$, we have $\mathrm{span}(v)$ is T-invariant. This means that if we take a basis for the eigenspace $E_\lambda(T)$, the span of each basis vector is a 1-dimensional T-invariant subspace. Thus we can write $E_\lambda(T)$ as a direct sum of 1-dimensional T-invariant subspaces.

If T is diagonalizable, recall this means we can write V as a direct sum of eigenspaces (we showed this in the finite dimensional case, but this generalizes), and hence V is a direct sum of 1-dimensional T-invariant subspaces. This explains the diagonal form of the matrix.

This also motivates a definition of **(internal) direct sum of linear operators**. Suppose $V = W_1 \oplus W_2$, and W_1, W_2 are T-invariant subspaces. Then $T|_{W_1}$ and $T|_{W_2}$ are well-defined. Then we say $T = T|_{W_1} \oplus T|_{W_2}$ and we have $(T|_{W_1} \oplus T|_{W_2})(w_1 + w_2) = T|_{W_1}(w_1) + T|_{W_2}(w_2)$.

Similarly the concept of an **external direct sum of linear operators** may be defined: Given V, W vector spaces, $T : V \to V$ and $S : W \to W$ linear, we define $T \oplus S : V \oplus W \to V \oplus W$ by $(T \oplus S)(v, w) = (T(v), T(w))$. (It should be clear how this can be extended to linear transformations where the domain and codomain are different.)

Then as we did with the internal and external direct sums of vector spaces, we can show there is a correspondence between the internal and external versions here. This makes it clearer why a decomposition of the vector space into invariant subspaces leads to a decomposition of the matrix of the linear operator into 1x1 blocks.

However, note that not all linear operators are diagonalizable (see example below). We will show later in the course that that we can (under the right basis) still decompose the matrix into blocks, they just may not be 1x1. That is to say, the invariant subspaces may have dimension larger than 1. This form of the matrix is called the **Jordan canonical form**.

Example 6.6

Consider $T : \mathbb{R}^3 \to \mathbb{R}^3$ defined by $x \mapsto Ax$ with

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Then we can check $T(e_1) = e_1, T(e_2) = e_1 + e_2, T(e_3) = e_3$. So we see $\text{span}(e_1)$ and $\text{span}(e_3)$ are invariant subspaces (eigenspaces), but $\text{span}(e_2)$ is not. However, we can check $\text{span}(e_1, e_2)$ is an invariant spaces.

Definition 6.7 (T-cyclic subspace of v) — Let $v \in V$. We define the ***T*-cyclic subspace of** v to be $W := \text{span}\{v, T(v), T^2(v), \dots\}$. This subspace is clearly T-invariant.

If additionally V is finite dimensional, then so is W (which implies only finitely many $T^k(v)$ form a linearly independent set).

Proposition 6.8

W is the smallest T-invariant subspace which contains v. (i.e. for any other T-invariant subspace W' which contains v, we have $W \subseteq W'$.)

Proof. Let W' be a T-invariant subspace containing v. Then by T invariance, we have $T(v) \in W'$, and inducting, we have $T^k(v) \in W'$ for all $k \in \mathbb{N}$. Hence $W \subseteq W'$. □

Then note $T|_W$ is well-defined. In particular we have

Theorem 6.9

Let V be finite dimensional and W be any invariant subspace of V. The characteristic polynomials of $T|_W$ and T satisfy $f_{T|_W}(t) \mid f_T(t)$.

Note that we have $V \simeq W \oplus V/W$, so we sort of want to do a decomposition of T with respect to these subspaces. For this we need to make sure the "restriction" of T to V/W is well-defined.

Proposition 6.10

Define $\overline{T} : V/W \to V/W$ by $\overline{T}(v + W) = T(v) + W$. If W is T-invariant this is well-defined.

Proof. Let $v_1 + W = v_2 + W$. Then $v_1 - v_2 \in W$, so $T(v_1) - T(v_2) \in T(W) \subseteq W$. Thus $T(v_1) + W = T(v_2) + W$. Hence we have $\overline{T}(v_1 + W) = T(v_1) + W = T(v_2) + W = \overline{T}(v_2 + W)$ as desired. $\square$

> **Remark 6.11.** This is not a literal restriction of T to V/W since we cannot literally consider V/W to be a subspace of V. This means $\overline{T}$ does behave a little differently than we'd expect, and I will explain more later.

Proof of Theorem 6.9. Let $\dim W = m$ and $\dim V = n$. Let $\mathcal{B}_W = \{v_1, \ldots, v_m\}$ be a basis for W. Extend this to a basis $\mathcal{B} = \{v_1, \ldots, v_n\}$ for V. Let $B = [T]_{\mathcal{B}_W}^{\mathcal{B}_W}$ and $A = [T]_{\mathcal{B}}^{\mathcal{B}}$.

Claim: $\overline{\mathcal{B}} = \{\overline{v}_{m+1}, \ldots, \overline{v}_n\}$ is a basis for V/W, where $\overline{v}_i := v_i + W$.

Pf: For any $v + W \in V/W$, write v as a linear combination $\sum_1^n a_i v_i$. Then $\sum_1^n a_i v_i + W = \sum_{m+1}^n a_i v_i + W$, since their difference $\sum_1^m a_i v_i \in W$. Then clearly $\overline{B}$ spans V/W, but they must also be linearly independent since $\{v_{m+1}, \ldots, v_n\}$ is linearly independent (exercise). $\square$

Now we write $B' = [\overline{T}]_{\overline{\mathcal{B}}}^{\overline{\mathcal{B}}}$.

Claim: We have

$$A = \left(\begin{array}{c|c} B & B'' \\ 0 & B' \end{array}\right)$$

Pf: We have

$$B = \begin{pmatrix} [T(v_1)]_{\mathcal{B}_W} & \ldots & [T(v_m)]_{\mathcal{B}_W} \end{pmatrix}$$

and

$$A = \begin{pmatrix} [T(v_1)]_{\mathcal{B}_W} & \ldots & [T(v_m)]_{\mathcal{B}_W} & [T(v_{m+1})]_{\mathcal{B}_W} & \ldots & [T(v_n)]_{\mathcal{B}_W} \end{pmatrix}$$

Fix $m + 1 \leq j \leq n$. Then we can write $T(v_j) = a_1 v_1 + \cdots + a_m v_m + a_{m+1} v_{m+1} + \cdots + a_n v_n$. Note that we have

$$\overline{T}(\overline{v}_j) = T(v_j) + W = \sum_{i=1}^n a_i v_i + W = \sum_{i=m+1}^n a_i v_i + W = \sum_{i=m+1}^n a_i (v_i + W) = \sum_{i=m+1}^n a_i \overline{v}_i$$

by the same arguments in the proof of the claim above.

In matrix form this is

$$[\overline{T}(\overline{v}_j)]_{\overline{\mathcal{B}}}^{\overline{\mathcal{B}}} = \begin{pmatrix} a_{m+1} \\ \vdots \\ a_n \end{pmatrix} \qquad \text{and} \qquad [T(v_j)]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} a_1 \\ \vdots \\ a_m \\ a_{m+1} \\ \vdots \\ a_n \end{pmatrix}$$

The form of the matrix A then follows. □

Now we can compute the characteristic polynomial:

$$\begin{aligned} f_T(t) &= f_A(t) \\ &= \det(A - tI_n) \\ &= \det \begin{pmatrix} B - tI_m & B'' \\ 0 & B' - tI_{n-m} \end{pmatrix} \\ &= \det(B - tI_m) \det(B' - tI_{n-m}) \\ &= f_{T|_W}(t) f_{\overline{T}}(t) \end{aligned}$$

Thus $f_{T|_W}(t) \mid f_T(t)$. □

Remark 6.12. Note that we have that $\overline{T}$ is a "restriction" of T in the sense that the characteristic polynomials multiply together, but if it were a true restriction, since we have $V \simeq W \oplus V/W$, we would expect the matrix of T to have a block diagonal form as explained earlier, but this is not the case.

We know that $V/W \simeq W'$ for some direct complement W' of W (which we can always explicitly construct). But if we look at W', we can see that it is not necessarily T-invariant, so $T|_{W'}$ would not be well-defined, and hence the matrix does not necessarily have block diagonal form.

This makes the direct complement harder to work with since for its matrix we would have to consider B'' along with B'. This is one of the advantages of using the quotient space V/W, since there is a natural to get a "restricted" operator $\overline{T}$.

Lecture 7 April 13, More Invariant Subspaces

> **Remark 7.1. Key Takeaway:** (from last time) If you have (or can write) a matrix in block upper triangular form, its characteristic polynomial is just the characteristic polynomials of the blocks on the diagonal multiplied together.

We do some examples:

Example 7.2

Consider $T : \mathbb{R}^3 \to \mathbb{R}^3$ defined by $(x, y, z) \mapsto (-y + z, x + z, 3z)$. The matrix of the operator in the standard basis is given by

$$A = \left(\begin{array}{cc|c} 0 & -1 & 1 \\ 1 & 0 & 1 \\ \hline 0 & 0 & 3 \end{array}\right)$$

Then we have $W = \operatorname{span}(e_1, e_2)$ is T-invariant, and $V/W = \operatorname{span}(\bar{e}_3) \simeq \operatorname{span}(e_3)$. Note we have $T(e_3) = e_1 + e_2 + 3e_3$. Quotienting out by W, we see $\overline{T}(\bar{e}_3) = 3\bar{e}_3$. (Since $T(e_3) + W = (e_1 + e_2 + 3e_3) + W = 3e_3 + W$.) Then

$$f_T(t) = f_A(t) = f_{T|_W}(t) f_{\overline{T}}(t) = (t^2 + 1)(3 - t)$$

Example 7.3

Consider $T : \mathbb{R}^4 \to \mathbb{R}^4$ defined by $x \mapsto Ax$ with

$$A = \left(\begin{array}{cc|cc} 1 & 1 & 2 & -1 \\ 0 & 1 & 0 & 1 \\ \hline 0 & 0 & 2 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right)$$

Then $W := \operatorname{span}(e_1, e_2)$ is T-invariant, and $V/W = \operatorname{span}(\bar{e}_3, \bar{e}_4)$. Then we see the matrix of $\overline{T}$ in this basis is the lower right block. Thus

$$f_T(t) = f_{T|_W}(t) f_{\overline{T}}(t) = \det \begin{pmatrix} 1 - t & 1 \\ 0 & 1 - t \end{pmatrix} \det \begin{pmatrix} 2 - t & -1 \\ 1 & 1 - t \end{pmatrix} = (1 - t)^2 (t^2 - 3t + 3)$$

Example 7.4

Consider $T : \mathbb{R}^4 \to \mathbb{R}^4$ defined by $x \mapsto Ax$ with

$$A = \left(\begin{array}{ccc|c} 2 & -1 & 0 & 0 \\ 1 & 1 & 1 & 2 \\ 3 & 5 & 4 & 1 \\ \hline 0 & 0 & 0 & 8 \end{array}\right)$$

We can try the same strategy as before, but we we would have to compute the determinant of a 3×3 matrix. As you can see, this is not very friendly to compute by hand, the larger the blocks get.

We give a result, which will give an alternate way to compute the characteristic polynomial in these cases (which I honestly don't think is easier), but will also lead to the important Cayley-Hamilton theorem.

Theorem 7.5

Let V be finite dimensional, $v \in V$, and W be the T-cyclic subspace of v.
($W = \operatorname{span}\{v, T(v), T^2(v), \dots\}$.) Let $\dim(W) = m$. Then

(1) $\mathcal{B} = \{v, T(v), \dots, T^{m-1}(v)\}$ is a basis of W. In particular $T^m(v)$ is a unique linear combination of $v, T(v), \dots, T^{m-1}(v)$.

(2) If $a_0 v + a_1 T(v) + \cdots + a_{m-1}T^{m-1}(v) + T^m(v) = 0$, then $f_{T|_W}(t) = (-1)^m(a_0 + a_1 t + \cdots + a_{m-1}t^{m-1} + t^m)$.

Proof of (1). Since $\#\mathcal{B} = \dim W$ it suffices to show $\mathcal{B}$ is linearly independent. Suppose not, so there exists $a_0, \dots, a_{m-1} \in F$ for which we have

$$a_0 v + a_1 T(v) + \cdots + a_{m-1}T^{m-1}(v) = 0$$

and at least one $a_i \neq 0$. Let j be the largest index for which $a_j \neq 0$. Then

$$a_0 v + a_1 T(v) + \cdots + a_j T^j(v) = 0$$

Then since $a_j \neq 0$, by dividing by a_j and rearranging, we see we can rewrite $T_j(v)$ as a linear combination of $\{v, T(v), \dots, T^{j-1}(v)\}$. Now $T^{j+1}(v) = T(T^j(v))$, so $T^{j+1}(v)$ can be written as a linear combination of $\{T(v), T^2(v), \dots, T^j(v)\}$, and by once again rewriting $T^j(v)$, this can reduced to a linear combination of $\{v, T(v), \dots, T^{j-1}(v)\}$. Inducting, we see every $T^k(v)$ can be written as such a linear combination, and hence $W \subseteq \operatorname{span}\{v, T(v), \dots, T^{j-1}(v)\}$. But $j \leq m-1$, so W is a spanned by a set of length $j \leq m - 1 < \dim(W)$, a contradiction. Thus W is linearly dependent, and we have proved the result. □

I believe this method is cleaner (see 115AH Worksheet 9):

Alternative Proof Sketch of (1). Since W is finite dimensional and any linearly independent set in W must have length at most the dimension, we can say there exists a largest $m \in \mathbb{N}$ such that $\mathcal{B} := \{v, T(v), \ldots, T^{m-1}(v)\}$ is linearly independent. Then $T^m(v)$ can be written as a linear combination of $\mathcal{B}$. We can then show span($\mathcal{B}$) is T-invariant and contains v, and since W is the smallest T-invariant subspace containing v, W must be contained in the span. The other direction of the inclusion is obvious, so we get $W = \text{span}\{v, T(v), \ldots, T^{m-1}(v)\}$ with $\dim(W) = m$ and $\mathcal{B}$ as a basis, as desired. □

Proof of (2). We have

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 0 & \cdots & 0 & -a_0 \\ 1 & & & -a_1 \\ & \ddots & & \vdots \\ & & 1 & -a_{n-1} \end{pmatrix}$$

(where the empty entries are presumed to be 0).

Then, inducting on n, we can compute the characeristic polynomial to be

$$f_T(t) = (-1)^m(a_0 + a_1 t + \cdots + a_{m-1}t^{m-1} + t^m)$$

□

Example 7.6

We return to the previous example:

Consider $T : \mathbb{R}^4 \to \mathbb{R}^4$ defined by $x \mapsto Ax$ with

$$A = \left(\begin{array}{ccc|c} 2 & -1 & 0 & 0 \\ 1 & 1 & 1 & 2 \\ 3 & 5 & 4 & 1 \\ \hline 0 & 0 & 0 & 8 \end{array}\right)$$

Take

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad T(e_1) = \begin{pmatrix} 2 \\ 1 \\ 3 \\ 0 \end{pmatrix}, \quad T^2(e_1) = \begin{pmatrix} 3 \\ 6 \\ 23 \\ 0 \end{pmatrix}, \quad T^3(e_1) = \begin{pmatrix} 0 \\ 26 \\ 131 \\ 0 \end{pmatrix}$$

(Note e_1 belongs to the invariant subspace corresponding to the 3×3 block.)

We can a solve a system of equations to write $T^3(e_1)$ as a linear combination of $\{e_1, T(e_1), T^2(e_1)\}$ (by the theorem above) and get $e_1 + 10T(e_1) - 7T^2(e_1) + T^3(e_1) = 0$. Then

$$f_{T|_W}(t) = (-1)^3(1 + 10t - 7t^2 + t^3) \implies f_T(t) = (-1)^3(1 + 10t - 7t^2 + t^3)(8 - t)$$

Remark 7.7. This requires you to solve a system of equations (or invert a matrix) so I really don't think it's computationally easier at all, unless you are lucky and able to solve the system by inspection. Of course, like I mentioned, the real value of the theorem we proved is that it leads to the Cayley-Hamilton theorem.

Lecture 8 April 15, Cayley-Hamilton Theorem

Theorem 8.1 (Cayley-Hamilton)

Let V be finite-dimensional and $T : V \to V$ be linear. Let $f_T(t) \in F[t]$ be the characteristic polynomial of T. Then $f_T(T) = 0$.

Similarly, if $A \in M_{n\times n}(F)$, and $f_A(t) \in F[t]$ is the characteristic polynomial of A, then $f_A(A) = 0$.

Remark 8.2. Here $F[t]$ denotes the set of formal polynomials with coefficients in F. That is, we do not treat the polynomials as functions but just as abstract expressions. $F[t]$ has a ring structure, and is often called the polynomial ring over F.

When plugging in a linear operator or matrix, we take the powers of the linear operator or matrix, so the zeroth power corresponds to the identity. That is, if $f(t) = c_0 + c_1 t + \dots c_n t^n$, then $f(T) = c_0 \operatorname{id}_V + c_1 T + \cdots + c_n T^n$.

Proof. Fix some $v \in V$ and consider its T-cyclic subspace $W = \operatorname{span}\{v, T(v), \dots\}$. Let $\dim(W) = m$, since V is finite-dimensional. Then by the theorem from last time we have there exists $a_0, \dots, a_{m-1} \in F$ such that

$$a_0 v + a_1 T(v) + \cdots + a_{m-1} T^{m-1}(v) + T^m(v) = 0.$$

Then since W is T-invariant, $T|_W$ is well-defined, and we have $f_{T|_W}(t) = (-1)^m(a_0 + a_1 t + \cdots + a_{m-1}t^{m-1} + t^m)$. Now since $f_{T|_W}(t) \mid f_T(t)$, we have $f_T(t) = g(t) f_{T|_W}(t)$ for some polynomial $g(t)$. Then $f_T(T)(v) = g(T)(f_{T|_W}(T)(v))$ but we know

$$f_{T|_W}(T)(v) = (a_0 \operatorname{id}_V + a_1 T + \cdots + a_{m-1}T^{m-1} + T^m)(v) = 0$$

Hence $f_T(T)(v) = 0$. Since this is true independent of v, we have $f_T(T) = 0$.

A similar proof for the matrix case follows. $\square$

Example 8.3 (Computing Powers of a Matrix)

The most annoying part of applying Cayley-Hamilton is calculating the powers of the matrix. However, in some cases, a clever decomposition makes it much easier.

We call a matrix A (and similar for linear operatr) **nilpotent** if there exists some $n \in \mathbb{N}$ for which $A^n = 0$.

Consider

$$A = \begin{pmatrix} a & 1 \\ 0 & a \end{pmatrix} = aI_2 + \underbrace{\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}}_{B}$$

Note that B is nilpotent, since $B^2 = 0$. Then we can compute by binomial expansion:

$$A^n = (aI_2 + B)^n = a^n I_2 + na^{n-1}B + \underbrace{\binom{n}{2} a^{n-2}B^2 + \cdots + B^n}_{\text{power of } B \geq 2} = a^n I_2 + na^{n-1}B$$

It is not too hard to show we can do this in higher dimensions, such as for

$$A = \begin{pmatrix} a & 1 & 0 \\ 0 & a & 1 \\ 0 & 0 & a \end{pmatrix} = aI_3 + \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

Remark 8.4. Note that in general it is easy to take powers of diagonal matrices since you just take power of each entry along the diagonal. Then we can see we can do a similar decomposition for any upper triangular matrix where we separate it into the diagonal and everything else above the diagonal. Try to convince yourself that if all nonzero entries lie above the diagonal, then the matrix is nilpotent (if you calculate some examples by hand, you will see the 0's sort of propagate across the diagonals as you repeatedly multiply).

However in fact, we can in general do this for any matrix because there exists something called the **Schur decomposition** which tells us we can put matrices (and linear operators) into upper triangular form.

As you can see, this makes taking polynomial functions of matrices much easier, but this also extends to infinite power series. This allows us to take functions of matrices as long as the functions are analytic, for example the exponential.

An Incorrect Proof

Professor Gao also gave the following as an example of an incorrect proof of the Cayley-Hamilton theorem:

Example 8.5 (Incorrect Proof of Cayley-Hamilton)

We have $f_A(A) = \det(A - AI_n) = \det(0) = 0$.

However, I do not believe he gave the correct explanation as to why the proof was incorrect, so I would like to spell it out in more detail.

The *actual* reason the proof does not work is because it confuses two different multiplication operations. The set of $n \times n$ matrices form what is called an **algebra**, that is, it is a vector space that additionally has a bilinear product $M_n(F) \times M_n(F) \to M_n(F)$ given by matrix multiplication.

The characteristic polynomial is given as $f_A(t) = \det(A - tI_n)$, and here tI_n is the scalar product. (It may also be useful to note that we are considering f_A as a function of t rather than a formal expression.) A priori it does not make sense to plug in a matrix $t = A$, because scalar products only work with elements of the underlying field F. The incorrect proof ignores this, and writes AI_n, which is the bilinear product (ordinary matrix multiplication) of the algebra. **These are not the same thing.**

At this point it's clear that the proof breaks logically.

> **Remark 8.6.** Everything below here are just my own ramblings. You can ignore it if you want.

But say we try to make sense of how to plug in $t = A$. We could attempt to understand the matrix's entries as being matrices themselves, so we consider them as elements of $M_n(M_n(F))$.

Professor Gao gives an example here:

Example 8.7

$$A = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$$

Let's try naively plugging in $t = A$ to the characteristic polynomial as a determinant. Then we have

$$f_A(t) = \det \begin{pmatrix} 2 - \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix} & 3 \\ 1 & 4 - \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix} \end{pmatrix} = 5 - 5I_2$$

Where we can clearly still take products between scalars and matrices, and combine like terms. And we see the result we get is "almost" 0. But the issue here is that when we plugged in $t = A$, we have $A * I_n$ (let's use that to denote this special product where the matrix itself is matrix-valued) is in $M_n(M_n(F))$, but the first A is still just in $M_n(F)$.

That suggests we should try to make sense of A as an element of $M_n(M_n(F))$, and one possible way to do it would be to attach I_2 matrices to each scalar entry, and we see that in that the case the characteristic polynomial really does evaluate to 0 (matrix).

It is not really strange to attach I_n to the scalars, because if you recall, when we evaluate the polynomial itself at a matrix, like $a_0 + a_1 t + \cdots + a_n t^n$, we attach the identity matrix to the constant term a_0 - this is the same idea.

Basically this example is **not** really a counterexample to the incorrect proof, but actually supports Cayley-Hamilton if you view it in the right way.

Lecture 9 April 17, Inner Product Spaces

This section mainly serves as review, so not every fact will be included. See 115AH notes for a more detailed treatment.

Let V be a vector space over F. For this section let $F = \mathbb{R}$ or $\mathbb{C}$.

Definition 9.1 (Inner Product) — An **inner product** on V is a map $\langle \cdot, \cdot \rangle : V \times V \to F$ such that

(i) (Linearity in first argument) $\langle ax + by, z\rangle = a\langle x, z\rangle + b\langle y, z\rangle$ for all $a, b \in F$ and $x, y, z \in V$.

(ii) (Conjugate symmetry) $\langle x, y\rangle = \overline{\langle y, x\rangle}$ for all $x, y \in V$.

(iii) (Positive definiteness) $\langle x, x\rangle \geq 0$ for all $x \in V$ and $\langle x, x\rangle = 0$ iff $x = 0$.

A vector space V equipped with an inner product $\langle \cdot, \cdot \rangle$ (written as a pair $(V, \langle \cdot, \cdot \rangle)$) is called an **inner product space**.

Proposition 9.2 (Antilinearity in second argument)
$\langle x, ay + bz\rangle = \overline{a}\langle x, y\rangle + \overline{b}\langle x, z\rangle$ for all $a, b, \in F$ and $x, y, z \in V$.

Remark 9.3. We often say the inner product is linear in the first argument and antilinear in the second. Other common terms include the inner product being Hermitian or sesquilinear.

Definition 9.4 (Norm) — A **norm** on V is a map $\|\cdot\| : V \to \mathbb{R}$ satisfying

(i) (Triangle inequality) $\|x + y\| \leq \|x\| + \|y\|$ for all $x, y \in V$.

(ii) (Absolute homoegeneity) $\|cx\| = |c|\|x\|$ for all $c \in F$ and $x \in V$.

(iii) (Positive definiteness) $\|x\| \geq 0$ for all $x \in V$ and $\|x\| = 0$ iff $x = 0$.

A **normed vector space** is a pair $(V, \|\cdot\|)$.

Proposition 9.5 (Inner products induce norms)
Given an inner product $\langle \cdot, \cdot \rangle$, it induces a natural norm defined by $\|x\| := \sqrt{\langle x, x\rangle}$

The proof that this is indeed a norm is left as an exercise.

Remark 9.6. Note that for norms, we explicitly restrict it to taking values in $\mathbb{R}$, instead of the field F the vector space V is defined over (since we want to measure lengths), and this is why we need the positive-definiteness of the inner product, and part of why we restrict ourselves to working with either $\mathbb{R}$ or $\mathbb{C}$.

Remark 9.7. In class we define the norm by the formula with the inner product, and then prove the axioms of norms from that, but defining it this way is more natural because not all norms are induced by inner products.

Example 9.8 (Standard Inner Products on $\mathbb{R}^n$ and $\mathbb{C}^n$)

The **standard inner product on** $\mathbb{R}^n$ is defined by

$$\langle (x_1, \dots, x_n), (y_1, \dots, y_n) \rangle = x_1 y_1 + \cdots + \cdots + x_n y_n$$

for all $(x_1, \dots, x_n), (y_1, \dots, y_n) \in \mathbb{R}^n$.

The **standard inner product on** $\mathbb{C}^n$ is defined by

$$\langle (x_1, \dots, x_n), (y_1, \dots, y_n) \rangle = x_1 \overline{y}_1 + \cdots + \cdots + x_n \overline{y}_n$$

for all $(x_1, \dots, x_n), (y_1, \dots, y_n) \in \mathbb{R}^n$.

Note that if we write $x = \begin{pmatrix} x_1 & \cdots & x_n \end{pmatrix}^T$ as a column vector and similarly for y, we have $\langle x, y \rangle = y^* x$ where y^* denotes the conjugate transpose of y.

The norms induced by both of these inner products are then just the standard Euclidean norms for $\mathbb{R}^n$ and $\mathbb{C}^n$.

Note $\langle \cdot, \cdot \rangle_{\mathbb{C}}|_{\mathbb{R}\times\mathbb{R}} = \langle \cdot, \cdot \rangle_{\mathbb{R}}$.

Theorem 9.9 (Cauchy-Schwarz Inequality)

For all $x, y \in V$ we have $|\langle x, y \rangle| \leq \|x\|\|y\|$. Equality holds iff x is a multiple of y.

Proof. Let $x, y \in V$. The inequality is trivially satisfied when $y = 0$, so assume $y \neq 0$. Motivated by the fact that equality occurs when x is a multiple of y, we look at

$$0 \leq \|x - cy\|^2 = \langle x - cy, x - cy \rangle = \|x\|^2 - \overline{c}\langle x, y \rangle - c\langle y, x \rangle + |c|^2 \|y\|^2$$

where $c \in F$ is arbitrary. Note that in the real case, the conjugation disappears, and the expression becomes a quadratic in c. We can treat $\|x - cy\|$ as the distance between the point x and the line $\operatorname{span}(y)$, and since equality occurs when x actually lies on this line, we should expect to recover the inequality by considering the worst case scenario - where the distance is minimized over c. This motivates us to consider $c = \langle x, y \rangle / \|y\|^2$. Plugging this in we get

$$\|x\|^2 - \frac{\overline{\langle x, y \rangle}}{\|y\|^2}\langle x, y \rangle - \frac{\langle x, y \rangle}{\|y\|^2}\langle y, x \rangle + \frac{|\langle x, y \rangle|^2}{\|y\|^2} \geq 0$$

Simplifying and multiplying by $\|y\|^2$ gives us

$$\|x\|^2 \|y\|^2 \geq |\langle x, y \rangle|^2$$

as desired. The condition for equality is obvious from this proof since we just take $x = cy$ for the particular value of c we chose. $\square$

Definition 9.10 (Angle) — Let $(V, \langle \cdot, \cdot \rangle)$ be a *real* inner product space. The angle $\theta(x, y)$ is the unique number in $[0, \pi)$ such that

$$\cos(\theta(x, y)) = \frac{\langle x, y \rangle}{\|x\| \|y\|}$$

Remark 9.11. Note that the Cauchy-Schwarz inequality guarantees the RHS has absolute value less than or equal to 1, so there exists some number whose cosine is equal to that, and hence this definition of angle is well defined.

You can also define angles for complex inner product spaces, but there is no single canonical choice. Common variations including taking the real part or the magnitude of the inner product.

Definition 9.12 (Orthogonality) — Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space. Let $x, y \in V$.

We say x, y are **orthogonal** and write $x \perp y$ if $\langle x, y \rangle = 0$.

We say a subset $S \subseteq V$ is **orthogonal** if the elements are pairwise orthogonal, i.e. if $v, u \in S$ and $v \neq u$, then v is orthogonal to u. We additionally say S is orthonormal if $\|v\| = 1$ for each $v \in S$.

Example 9.13

For $V = \mathbb{R}^n$, $\langle \cdot, \cdot \rangle$ the standard inner product, the standard basis $\{e_1, \dots, e_n\}$ is orthonormal.

For $V = \mathbb{R}^3$ with the standard inner product,

$$S = \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} \right\} \text{ is an orthogonal basis}$$

$$S' = \left\{ \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{6}} \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} \right\} \text{ is an orthonormal basis}$$

Proposition 9.14

Any orthogonal set $S \subseteq V \setminus \{0\}$ is linearly independent.

Proof. Suppose $c_1 v_1 + \dots + c_k v_k = 0$ is a linear combiantion of elements in S. Then for each i

$$0 = \langle 0, v_i \rangle = \langle c_1 v_1 + \dots + c_k v_k, v_i \rangle = c_1 \langle v_1, v_i \rangle + \dots + c_k \langle v_k, v_i \rangle = c_i \langle v_i, v_i \rangle = c_i$$

Then we must have each $c_i = 0$, and hence S is linearly independent. □

This means when we want to show a set is an orthogonal basis, it suffices to check orthogonality (which is usually easier than linear independence) and spanning. But this proof also suggests the inner product is really useful for extracting coefficients out of linear combinations when given an orthogonal basis. In particular we have:

Proposition 9.15

Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space and $\mathcal{B}$ be an orthogonal basis. Then for all $x \in V$ we have

$$x = \sum_{i=1}^{n} \frac{\langle x, v_i\rangle}{\|v_i\|^2} v_i$$

Proof. Since $\mathcal{B}$ is a basis, for any $x \in V$ we can write $x = \sum_{i=1}^{n} c_i v_i$ for $c_i \in F$. Then for each j

$$\langle x, v_i\rangle = \sum_{j=1}^{n} c_j \langle v_j, v_i\rangle = c_i\langle v_i, v_i\rangle = c_i\|v_i\|^2$$

Rearranging for c_i, the formula follows. □

Having an explicit formula to extra the coefficients of a linear combination actually allows us to construct an orthogonal basis from any *finite* basis.

Algorithm 9.16 (Gram-Schmidt) — The idea is as follows: Start with a basis $\{w_1, \dots, w_n\}$ of V. We take the first vector as the first vector of our orthogonal basis. Take the second vector and remove the "part" which is parallel to w_1, so we remain with something that is orthogonal to w_1. Then take the third vector and remove the "parts" which are parallel to the first two vectors in our orthogonal basis, and repeat this process until we have finished.

- Start with a basis $\{w_1, \dots, w_n\}$ of V.
- Set $v_1 := w_1$.
- Set $v_2 := w_2 - \dfrac{\langle w_2, v_1\rangle}{\|v_1\|^2} v_1$
- Set $v_j := w_j - \displaystyle\sum_{i=1}^{j-1} \frac{\langle w_j, v_i\rangle}{\|v_i\|^2} v_i$

Then $\{v_1, \dots, v_n\}$ is an orthogonal basis for V. Normalizing, we get an orthonormal basis of V:

$$\left\{\frac{v_1}{\|v_1\|}, \dots, \frac{v_n}{\|v_n\|}\right\}$$

Having an orthonormal basis also makes computing the matrix of a finite dimensional linear operator much easier, and there is good reason for this.

Remark 9.17. Let V be a finite dimensional inner product space. Given an orthonormal basis $\{v_1, \dots, v_n\}$, let $x, y \in V$. We can write $x = \sum_{i=1}^{n} x_i v_i$ and $y = \sum_{i=1}^{n} y_i v_i$, and we can check $\langle x, y\rangle = x_1y_1 + \cdots + x_ny_n$. So in an orthonormal basis, the inner product essentially reduces to the standard inner product over F ($\mathbb{R}$ or $\mathbb{C}$). Since finite dimensional inner product spaces always have an orthonormal basis (by Gram-Schmidt), this sort of means that up to choosing a suitable basis, the "only" inner product on finite dimensional inner product spaces is the standard one.

This very much mirrors how the matrix and vector multiplication is like the standard inner product on the standard basis. In particular, we have:

Proposition 9.18

Let $(V, \langle \cdot, \cdot \rangle)$ be a finite dimensional inner product space, and let $\dim V = n$ and $\mathcal{B}$ be an orthonormal basis. Let $T : V \to V$ be a linear operator. Let a_{ij} denote the entry in the ith row and the jth column of the matrix $[T]_{\mathcal{B}}^{\mathcal{B}}$. Then $a_{ij} = \langle T(v_j), v_i \rangle$ for all i, j.

Proof. We have
$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} [T(v_1)]_{\mathcal{B}} & \cdots & [T(v_n)]_{\mathcal{B}} \end{pmatrix}$$
But since we have an orthonormal basis we must have $T(v_j) = \sum_{i=1}^{n} \langle T(v_j), v_i \rangle v_i$ by Proposition 9.15. Since $T(v_j)$ gives the jth column, we see the ith row entry is exactly $\langle T(v_j), v_i \rangle$. □

Definition 9.19 (Orthogonal Complement) — Let $S \subseteq V$. Then we define the **orthogonal complement** of S as $S^\perp := \{x \in V : \langle x, y \rangle = 0 \quad \forall\, y \in S\}$. Note $S^\perp$ is always a vector space even if S is not.

Theorem 9.20

Let V be a finite dimensional inner product space and W be a subspace. Then

(i) $V = W \oplus W^\perp$. That is, for all $v \in V$, there exists unique $u \in W$ and $z \in W^\perp$ such that $v = u + z$.

(ii) $\dim V = \dim W + \dim W^\perp$.

There are some weaker statements you can make when V is not finite dimensional, see 115AH notes for details. The first part can be proven by taking a basis for W and $W^\perp$, and the second part follows immediately from properties of the direct sum.

Theorem 9.21

Let V be a finite dimensional inner product space and $S = \{v_1, \ldots, v_k\} \subseteq V$ be orthonormal.

(i) S can be extended to an orthonormal basis of V given by $\{v_1, \ldots, v_k, v_{k+1}, \ldots, v_n\}$.

(ii) $\{v_{k+1}, \ldots, v_n\}$ is a basis for $S^\perp$.

The first statement can be proven by first extending S to any kind of basis, and then using Gram-Schmidt (which leaves the original vectors unchanged). The second statement then immediately follows.

Remark 9.22. Note that the first statement fails spectacularly in infinite dimensional spaces. **NO** (Hamel) basis of an infinite dimensional space can be orthogonal. A **Hamel basis** is just a basis in the usual sense, where you can only have finite linear combinations. This is to distinguish from a **Schauder basis**, which allows infinite linear combinations. You may hear that the set $\{e^{2\pi int} : n \in \mathbb{Z}\}$ is an orthonormal basis in Fourier series, but it is a basis in the Schauder sense, not the Hamel sense.

Suppose you had an orthonormal (Hamel) basis for a Hilbert space (a complete inner product space, just so we can make sure limits exist). Take some countable subset $\{v_1, v_2, \dots\}$ of the basis. Then consider the infinite sum

$$v := \sum_{i=1}^{\infty} 2^{-i} v_i$$

which converges (we can talk about convergence since there is a natural norm topology induced by the inner product). But we have a Hamel basis it must also be able to be written as finite linear combination $v = a_1 x_1 + \cdots + a_n x_n$. Then we can extract coefficients by taking

$$2^{-k} = \langle v, v_k \rangle = \sum_{j=1}^{n} a_j \langle x_j, v_k \rangle$$

So there must be some x_j and v_k which are not orthogonal, a contradiction.

Intuitively the reason why this fails is that the inner product gives us this super rigid structure for extracting coefficients of linear combinations of basis elements, but (Hamel) bases in infinite dimensional spaces are usually too large for this to be self-consistent.

Lecture 10 April 20, Adjoints and Some Spectral Theory

Unless otherwise specified, let $(V, \langle \cdot , \cdot \rangle_V)$ and $(W, \langle \cdot , \cdot \rangle_W)$ be inner product spaces over F ($\mathbb{R}$ or $\mathbb{C}$).

Definition 10.1 (Adjoint) — Let $T : V \to W$ be linear. If there exists a linear map $T^* : W \to V$ such that

$$\langle T(x), y\rangle_W = \langle x, T^*(y)\rangle_V \quad \text{for all } x \in V \text{ and } y \in W$$

then T^* is called the **adjoint** of T.

Given an $m \times n$ matrix A, its **adjoint** is defined as $A^* = (\overline{A})^T$.

Remark 10.2. We omit most of the basic properties of the adjoint and assume them without proof. See 115AH notes for more detail.

Proposition 10.3

Suppose V, W are finite dimensional, and let $\mathcal{B}$ and $\mathcal{C}$ be orthonormal bases for V and W, respectively. Then

$$[T^*]_{\mathcal{B}}^{\mathcal{C}} = ([T]_{\mathcal{C}}^{\mathcal{B}})^*$$

See 115AH notes for the explicit computation.

Theorem 10.4

If V is finite dimensional and $T : V \to W$ is linear, then there exists a unique adjoint $T^* : W \to V$

This proof requires some dual spaces, which should be more familiar now! It first proves a finite dimensional version of the Riesz Representation theorem, and uses that to prove the result. See 115AH notes for details.

Remark 10.5. Note that Professor Gao defines the adjoint for linear operators via the matrix definition, and then derives the inner product property. This version generalizes to infinite dimensional spaces, although adjoints may not always exist in infinite dimensional spaces. (They are very important to functional analysis and PDE's!)

We include the proof for the opposite direction for the sake of completeness. (We only look at the linear operator case for simplicity, but it can be easily extended to linear maps $T : V \to W$.)

Theorem 10.6

Let V be finite dimensional and $\mathcal{B} = \{v_1, \ldots, v_n\}$ be an orthonormal basis. Let $T : V \to V$ be a linear operator and $A := [T]_{\mathcal{B}}^{\mathcal{B}}$. Define $T^* : V \to V$ such that $[T^*]_{\mathcal{B}}^{\mathcal{B}} = A^*$. Then $\langle T(x), y\rangle = \langle x, T^*(y)\rangle$.

Proof. By linearity it suffices to show this for x, y in the orthonormal basis. That is, we want to show $\langle T(v_i), v_j\rangle = \langle v_i, T(v_j)\rangle$ for all i, j.

Let a_{ji} denote the entry in the jth row and ith column of A. We have $\langle T(v_i), v_j\rangle = a_{ji}$. Note $\langle v_i, T^*(v_j)\rangle = \overline{\langle T^*(v_j), v_i\rangle}$. But the expression inside the conjugation is the entry in the ith row and the jth column of A^*. But by the definition of A^* that is exactly $\overline{a_{ji}}$. It then follows $\langle T(v_i), v_j\rangle = \langle v_i, T^*(v_j)\rangle$, and we are done. $\square$

Proposition 10.7

Let V be finite dimensional and $T : V \to V$ be a linear operator. λ is an an eigenvalue for T iff $\overline{\lambda}$ is an eigenvalue of T^*.

Proof. Recall that $\det A^T = \det A$. Let A be the matrix of T in some basis. Then

$$\det(A - \lambda I) = 0 \iff \det\left(\overline{A - \lambda I}\right)^T = 0 \iff \det((\overline{A})^T - \overline{\lambda} I) = 0 \iff \det(A^* - \overline{\lambda} I) = 0$$

$\square$

Remark 10.8. Clearly this proof fails in infinite dimensions, but the result also just fails because failure of invertibility does not always imply failure of injectivity (this was true in finite dimensions because of Rank-Nullity).

Theorem 10.9

Let V be any finite dimensional vector space (not necessarily an inner product space) and $T : V \to V$ be a linear operator. Then there exists a basis of V for which the matrix of T is upper triangular iff the characteristic polynomial of T splits.

Proof. Let $\dim V = n$.

($\implies$) Suppose $\mathcal{B}$ is a basis of V such that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is upper triangular. Then from basic determinant properties we have the characteristic polynomial of T is given by $f_T(t) = (a_1 - t)(a_2 - t) \cdot (a_n - t)$ where a_i are the diagonal entries. Then clearly the polynomial splits.

($\impliedby$) Suppose the characteristic polynomial of T splits. Let us induct on the dimension n. If $n = 1$, then any matrix is upper triangular, so the base case is trivially true.

Suppose the result is true for any operator in $n-1$ dimensions, and consider T in n dimensions. Since the characteristic polynomial splits, there exists an eigenvalue λ_1 and an eigenvector v_1. Then $\mathrm{span}(v_1)$ is an invariant subspace. Then the canonical restriction to the quotient $\overline{T} : V/\mathrm{span}(v_1) \to V/\mathrm{span}(v_1)$ is well-defined, and we have $\dim(V/\mathrm{span}(v_1)) = n-1$. (This is because the projection map $\pi : V \to V/\mathrm{span}(v_1)$ has kernel $\mathrm{span}(v_1)$ and then apply the First Isomorphism theorem.)

Now $T|_{\mathrm{span}(v_1)}$ is well-defined, and recall its characteristic polynomial not only divides the characteristic polynomial of T, but we actually have

$$f_T(t) = f_{T|_{\mathrm{span}(v_1)}}(t) f_{\overline{T}}(t)$$

(See the proof of Theorem 6.9.) Thus the characteristic polynomial of $\overline{T}$ must also split over F, and by the inductive hypothesis, there exists some basis $\{\overline{v}_2, \ldots, \overline{v}_n\}$ such that the matrix of $\overline{T}$ is upper triangular. Note that by taking $\{v_2, \ldots, v_n\}$ in V such that $\pi(v_i) = \overline{v_i}$ (the quotient map), we get a full basis $\{v_1, \ldots, v_n\}$ of V. In the proof of Theorem 6.9, we then showed the matrix of T has the block upper triangular form

$$A = \left(\begin{array}{c|c} B & B'' \\ \hline 0 & B' \end{array}\right)$$

where B is the matrix of $\mathrm{span}(v_1)$ and B' is the matrix of $\overline{T}$, in the respective bases we have defined. Since B and B' are upper triangular, it follows A is upper triangular. $\square$

Corollary 10.10

If V is a finite dimensional *complex* vector space (so $F = \mathbb{C}$), then there exists a basis for which the matrix of T is upper triangular.

Proof. This follows from the Fundamental Theorem of Algebra, which says every polynomial in $\mathbb{C}$ splits. $\square$

Theorem 10.11 (Schur Decomposition)

If V is a finite dimensional complex inner product space, then there exists an *orthonormal* basis for which the matrix of T is upper triangular.

Proof. This proof requires the following observation about the Gram-Schmidt process: Let $\{w_1, \ldots, w_n\}$ be the basis we are trying to orthonormalize, and at each step of the process, we have the orthonormal set $\{v_1, \ldots, v_k\}$. Then $\mathrm{span}(w_1, \ldots, w_k) = \mathrm{span}(v_1, \ldots, v_k)$. In other words, the Gram-Schmidt process preserves the span at each step.

Now take a basis $\mathcal{B} = \{v_1, \ldots, v_n\}$ such that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is upper triangular. Note that this is equivalent to $T(v_i) \in \mathrm{span}(v_1, \ldots, v_k)$ for each k and $i \leq k$ (draw the matrix out to convince yourself). Perform Gram-Schmidt and normalize to get an orthonormal basis $\mathcal{B}' = \{w_1, \ldots, w_n\}$. But note that in Gram-Schmidt each v_i is a linear combination of $\{w_1, \ldots, w_i\}$. Then it follows that for each k and $i \leq k$ we have $T(v_i) \in \mathrm{span}(w_1, \ldots, w_k) = \mathrm{span}(v_1, \ldots, v_k)$, so the matrix of T in the orthonormal basis $\mathcal{B}'$ is upper triangular. $\square$

Definition 10.12 (Normal and Self-Adjoint Operators) — A linear operator $T : V \to V$ is called **normal** if $TT^* = T^*T$. T is called **self-adjoint** if $T = T^*$. These concepts are similarly defined for matrices.

Proposition 10.13
Assume T is normal. Then

(i) $\|T(x)\| = \|T^*(x)\|$ for all $x \in V$.

(ii) $T - cI$ is normal.

(iii) $x \in V$ is an eigenvector of T with eigenvalue λ iff it is an eigenvector of T^* with eigenvalue $\overline{\lambda}$ of T^*.

(iv) If $\lambda_1 \neq \lambda_2$ and λ_1 is an eigenvalue of x_1 and λ_2 an eigenvalue of x_2, then $\langle x_1, x_2 \rangle = 0$.

Proof. We have

(i) $\|T(x)\|^2 = \langle T(x), T(x)\rangle = \langle x, T^*(T(x))\rangle = \langle x, T(T^*(x))\rangle = \langle T^*(x), T^*(x)\rangle = \|T^*(x)\|^2$.

(ii) $(T - cI)(T - cI)^* = TT^* - cT - \overline{c}T^* + |c|^2 = T^*T - \overline{c}T^* - cT + |c|^2 = (T - cI)^*(T - cI)$.

(iii) Let $U = T - \lambda I$, which is normal. Then $U(x) = 0 \iff \|U(x)\| = 0 \iff \|U^*(x)\| = 0 \iff U^*(x) = 0$. But $U(x) = 0$ means x is an eigenvector of T with eigenvalue λ.

(iv) $\langle T(x_1), x_2\rangle = \langle x_1, T^*(x_2)\rangle \implies \langle \lambda_1 x_1, x_2\rangle = \langle x_1, \overline{\lambda}_2 x_2\rangle \implies (\lambda_1 - \lambda_2)\langle x_1, x_2\rangle = 0$. Since $\lambda_1 \neq \lambda_2$, we must have $\langle x_1, x_2\rangle = 0$.

□

Proposition 10.14
Let T be self-adjoint. Then

(i) All eigenvalues are real.

(ii) Let V be finite dimensional. The characteristic polynomial splits over F ($\mathbb{R}$ or $\mathbb{C}$).

Proof. (i) Note that if T is self-adjoint, then it is also normal. Then let λ be an eigenvalue of T and x be a corresponding eigenvector. Then $T(x) = \lambda x$ and $T^*(x) = \overline{\lambda}x$, so $\lambda x = \overline{\lambda}x$, so $\lambda = \overline{\lambda}$, and $\lambda \in \mathbb{R}$.

(ii) This is trivially true for $F = \mathbb{C}$ due to the Fundamental Theorem of Algebra.

Let $F = \mathbb{R}$. Pick a basis $\mathcal{B}$ such that $A = [T]_{\mathcal{B}}^{\mathcal{B}}$ is upper triangular. Since T is self-adjoint, we have $A = A^*$ as well, and since A^* is lower triangular, it follows A is diagonal. The diagonal then contains its eigenvalues, which must be real, so its characteristic polynomial splits over $\mathbb{R}$.

□

Theorem 10.15 (Spectral Theorem for Finite Dimensional Spaces, Part I)
We have the following:

(i) Let $F = \mathbb{C}$. T is orthogonally diagonalizable (there exists an orthonormal eigenbasis iff T is normal.

(ii) Let $F = \mathbb{R}$. T is orthogonally diagonalizable (there exists an orthonormal eigenbasis iff T is self-adjoint.

For the proof see 115AH notes.

Remark 10.16. This is the probably the most important result for inner product spaces. We get an extremely strong result on diagonalizability, and furthermore we get an orthonormal basis. This does not extend well into infinite dimensions for two reasons. One is that the proof relies on induction on the dimension.

And the other is that in infinite dimensions it is not true that injectivity $\iff$ surjectivity $\iff$ inveritiblity for linear operators, so a fundamental idea, that an operator has an eigenvalue when $T - \lambda \mathbf{1}_V$ is not invertible, also breaks down. So the spectrum of an operator (sort of like the set of eigenvalues) is also defined differently.

If you are interested in this, considering taking a course in functional analysis.

Lecture 11 April 22, Unitary and Orthogonal Operators

Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space over F.

Remark 11.1. Note that if we have an inner product space, as we said, there is a natural norm induced by $\|x\| = \sqrt{\langle x, x\rangle}$. But if have the norm induced by the inner product, can we recover the inner product from the induced norm? Yes, but we have different formulas depending on the underlying field:

- For $F = \mathbb{R}$, we have
$$\langle x, y\rangle = \frac{1}{4}\left(\|x+y\|^2 - \|x-y\|^2\right)$$
- For $F = \mathbb{C}$, we have
$$\langle x, y\rangle = \frac{1}{4}\left(\|x+y\|^2 - \|x-y\|^2\right) + \frac{1}{4}i\left(\|x+iy\|^2 - \|x-iy\|^2\right)$$

Remark 11.2. Note that for inner product spaces we have the **parallelogram identity**:
$$\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2$$
Then the above identity for $F = \mathbb{R}$ can also be rewritten in the form Professor Gao gave:
$$\langle x, y\rangle = \frac{1}{2}\left(\|x+y\|^2 - \|x\|^2 - \|y\|^2\right)$$

Remark 11.3. Note this does **NOT** mean that every norm comes from an inner product. This is not true, and inner product spaces are a strict subclass of normed vector spaces. For an example, take the taxicab norm (or the 1-norm) on $\mathbb{R}^2$ defined by $\|(x, y)\|_1 = |x| + |y|$, and check the "inner product" defined by the above formula is not linear in each term.

The above identity **ONLY** works for the norm that is induced from the given inner product. In fact you can show a norm is induced by an inner product iff it satisfies the parallelogram identity.

This close relationship between the inner product and the induced norms suggests operators which preserve one might preserve the other.

Definition 11.4 (Orthogonal and Unitary Operators) — Let $T : V \to V$ be a linear operator such that $\langle T(x), T(y)\rangle = \langle x, y\rangle$ for all $x, y \in V$. If $F = \mathbb{R}$ we say T is **orthogonal**, and if $F = \mathbb{C}$ we say T is **unitary**. Similar notions are defined for matrices.

Let us prove some equivalent characterizations for orthogonal/unitary operators which motivate the naming.

Theorem 11.5 (Characterization of Orthogonal/Unitary Operators)
Let $T : V \to V$ be a linear operator. The following are equivalent:

(i) $T^*T = \mathbf{1}_V$

(ii) $TT^* = \mathbf{1}_V$

(iii) $\langle T(x), T(y)\rangle = \langle x, y\rangle$ for all $x, y \in V$.

If additionally V is finite dimensional, following are also equivalent:

(iv) If $\mathcal{B}$ is a (finite) orthornormal basis of V, then so is $T(\mathcal{B})$.

(v) There exists some orthonormal basis $\mathcal{B}$ of V such that $T(\mathcal{B})$ is an orthonormal basis as well.

(vi) $\|T(x)\| = \|x\|$ for all $x \in V$.

Proof. (i) $\iff$ (ii) Recall that if an operator on a finite dimensional space (or matrix) is left or right invertible, then it is automatically invertible (both left and right). This equivalence then follows immediately.

(i) $\implies$ (iii) For all $x, y \in V$ we have $\langle T(x), T(y)\rangle = \langle x, T^*(T(y))\rangle = \langle x, y\rangle$.

(iii) $\implies$ (i) Fix any $y \in V$. Then for all $x \in V$ we have $\langle x, y\rangle = \langle T(x), T(y)\rangle = \langle x, T^*(T(y))\rangle$. Thus $y = T^*(T(y))$ for each $y \in V$, and hence $T^*T = \mathbf{1}_V$.

Now for the finite dimensional equivalences:

(iii) $\implies$ (iv) Let $\mathcal{B} = \{v_1, \ldots, v_n\}$ be an orthonormal basis. Then $\langle v_i, v_j\rangle = \delta_{ij}$ But $\langle T(v_i), T(v_j)\rangle = \langle v_i, v_j\rangle$, so we have $\{T(v_1), \ldots, T(v_n)\}$ is an orthonormal basis as well.

(iv) $\implies$ (v) Trivial.

(v) $\implies$ (vi) Let $\mathcal{B} = \{v_1, \ldots, v_n\}\}$ be a (finite) orthonormal basis, so that $T(\mathcal{B})$ is as well. Then recall the Pythagorean theorem, which says if x and y are orthogonal, then $\|x + y\|^2 = \|x\|^2 + \|y\|^2$. We can easily extend this by induction to sums of n pairwise orthogonal vectors. Then since for any $x \in V$ we can write $x = \sum_1^n a_i v_i$, we have

$$\|x\|^2 = \sum_{i=1}^n |a_i|^2 \|v_i\|^2 = \sum_{i=1}^n |a_i|^2 = \sum_{i=1}^n |a_i|^2 \|T(v_i)\|^2 = \left\| \sum_{i=1}^n a_i T(v_i) \right\| = \|T(x)\|^2$$

(vi) $\implies$ (iii) Using the formulas in Remark 11.1, we see that preserving norm, will also preserve the inner product. As an example for when $F = \mathbb{R}$:

$$\langle T(x), T(y)\rangle = \frac{1}{4}\left(\|T(x) + T(y)\|^2 - \|T(x) - T(y)\|^2\right) = \frac{1}{4}\left(\|x + y\|^2 - \|x - y\|^2\right) = \langle x, y\rangle$$

The complex case follows similarly.

□

Corollary 11.6

An $n \times n$ matrix A is unitary iff its column vectors form an orthonormal basis.

Proof. Consider (iii) $\implies$ (iv) and choose the standard basis of $\mathbb{R}^n$. $\square$

Remark 11.7. Recall in Remark 9.22 we showed that no basis in an infinite dimensional space can be orthonormal, so we can safely assume it is finite here. Like all the eigenstuff, orthonormal bases isn't one of the things that translate well to infinite dimensions.

One of the dilemmas of writing these recent sections is how to treat the infinite dimensional case, since a lot of these concepts are defined differently in the infinite-dimensional case. For example, I avoid defining unitary operators as isometries (see below) because the equivalence (iii) $\iff$ (v) is not true in infinite dimensions. In particular there are isometries which are not unitary operators.

However, unitary operators in infinite dimensional spaces are defined a little differently (they are required to additionally be surjective and a bounded operator), but the equivalences of (i)-(iii) still hold. So the definition I chose still has some hope of working by adding some extra conditions, but be careful about directly applying arguments to the infinite dimensional case.

The reason we need these extra conditions is that for infinite dimensional spaces (particularly those that appear in functional analysis), we often have (and need) some topology on the spaces so we can have some reasonably strong results, and these definitions will naturally interact with the topology.

Definition 11.8 (Isometry) — Let $(V, \|\cdot\|)$ be a normed vector space, and $T : V \to V$ be a linear operator. T is called an **isometry** if $\|T(x)\| = \|x\|$ for all $x \in V$.

Remark 11.9. As you may guess by the same, the name "isometry" actually means "same distance," so it should really be defined on metric spaces, but norms induce a natural metric $d(x, y) = \|x - y\|$. We see that an isometry as defined for normed spaces is the same as saying $d(T(x), T(y)) = d(x, y)$.

This tells us that the unitary operators (on finite dimensional spaces) are exactly the isometries, which should not be surprising considering we showed that we can recover an induced norm from an inner product and vice versa.

Remark 11.10. Note that since orthogonal operators (on finite dimensional spaces) preserve both lengths (norm) and inner products, they actually preserve angles. If $F = \mathbb{R}$, we have

$$\cos(\theta(x, y)) = \frac{\langle x, y\rangle}{\|x\|\|y\|} = \frac{\langle T(x), T(y)\rangle}{\|T(x)\|\|T(y)\|}$$

In this way they are named "orthogonal" operators, because they preserve the rigidity of the space. (For example in $\mathbb{R}^n$ you can think of them as preserving the coordinate grid, and you can show that orthogonal operators consist of compositions of only rotations and reflections).

Lecture 12 April 24, More Unitary and Orthogonal Operators

Example 12.1

We give some examples of orthogonal and unitary matrices:

1. $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ is orthogonal.
2. $\begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}$ is unitary
3. $\frac{1}{2}\begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix}$ is unitary

Example 12.2

Let L be a line through the origin in $\mathbb{R}^2$ (i.e. a 1-dimensional subspace). Define $T : \mathbb{R}^2 \to \mathbb{R}^2$ by

$$T(x) = \begin{cases} x & x \in L \\ -x & x \in L^\perp \end{cases}$$

Since $\mathbb{R}^2 = L \oplus L^\perp$, extend this by linearity to a function on all of T. Note that T can be interepreted as reflection in the line L.

We can compute the matrix of T by taking the standard basis, and this gives us some matrix A, but by taking the appropriate basis, the matrix becomes very simple. If we let α be the angle between L and the positive x-axis, note $(\cos\alpha, \sin\alpha) \in L$ and $(-\sin\alpha, \cos\alpha) \in L^\perp$, and they form an orthonormal basis. In this basis the matrix becomes

$$B = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

But A and B represent the same transformation, and hence must be similar. They are related by the change of basis matrix from our orthonormal basis to the standard basis given by

$$Q = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix}$$

so $A = QBQ^{-1}$. Note that Q is orthogonal since its columns are the orthonormal basis. Since Q is rotation CCW by α, this decomposition of A can be interpreted as rotating by $-\alpha$, so L becomes the x-axis, apply B, which reflects across the x-axis, and rotate back by α, effectively achieving reflection in L.

This motivates the following definition:

Definition 12.3 (Orthogonally/Unitarily Equivalent) — Let $A, B \in M_{n\times n}(F)$. We say

- If $F = \mathbb{R}$, A and B are **orthogonally equivalent** if $A = QBQ^{-1}$ for some orthogonal matrix Q.
- If $F = \mathbb{C}$, A and B are **unitarily equivalent** if $A = QBQ^{-1}$ for some unitary matrix Q.

Remark 12.4. Note that orthogonal matrices are defined to be real, so we must check that Q not only satisfies $Q^TQ = I$ but also it has all real entries. It is possible for A, B to be real, $Q^TQ = I$, and Q to have complex entries.

Now we can restate Schur Decomposition for matrices:

Theorem 12.5 (Schur Decomposition for Matrices)
Let $A \in M_{n\times n}(F)$. Then A is orthogonally/unitarily equivalent to an upper triangular matrix iff the characteristic polynomial of A splits over F.

Note this also lets us restate the spectral theorem for matrices. Being orthogonally diagonalizable exactly means having an orthonormal basis for which the matrix representation is diagonal, and the hence the change of basis matrix is orthogonal/unitary.

Definition 12.6 — Let $A \in M_{n\times n}(F)$. Recall $A^* := \overline{A}^T$. We say A is

- **symmetric** if $F = \mathbb{R}$ and $A = A^T$.
- **Hermitian** if $F = \mathbb{C}$ and $A = A^*$.
- **normal** if $F = \mathbb{C}$ and $AA^* = A^*A$,

Theorem 12.7 (Spectral Theorem for Matrices)
We have

1. Let $A \in M_{n\times n}(\mathbb{R})$. Then A is orthogonally equivalent to a diagonal matrix iff A is symmetric.
2. Let $A \in M_{n\times n}(\mathbb{C})$. Then A is unitarily equivalent to a diagonal matrix iff A is normal, and A is Hermitian iff all the eigenvalues are real.

This is immediate given a proof of the spectral theorem for linear operators, but since Professor Gao gave a proof in the language of orthogonal/unitary matrices, we include it here for completeness:

Proof. 1. ($\Longrightarrow$) For some diagonal D and orthogonal Q we have $A = Q^{-1}DQ = Q^TDQ$. Thus $A^T = Q^TD^T(Q^T)^T = Q^TDQ = A$ and A is symmetric.

($\Longleftarrow$) Being symmetric is equivalent to being self-adjoint over $\mathbb{C}$, meaning all eigenvalues are real, and hence the characteristic polynomial splits over $\mathbb{R}$. Then by Schur decomposition $A = Q^{-1}UQ$ for some orthogonal Q and upper triangular U. Then $U = QAQ^{-1} = QAQ^T$, and thus $U^T = (Q^T)^T A^T Q^T = QAQ^T = U$, and since U is upper triangular, U^T is lower triangular, so U must be diagonal, as desired.

2. ($\Longrightarrow$) For some diagonal D and unitary Q we have $A = Q^{-1}DQ = Q^*DQ$. Then it is easy to check $A^*A = AA^*$ via computation.

   ($\Longleftarrow$) Consider $T : \mathbb{C}^n \to \mathbb{C}^n$ defined by $T(x) = Ax$. Since polynomials over $\mathbb{C}$ always split, by Schur Decomposition there exists some orthonormal basis $\mathcal{B} = \{v_1, \ldots, v_n\}$ such that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is upper triangular. If we denote the entry in the ith row and jth column by a_{ij}, we have

$$\begin{aligned} T(v_1) &= a_{11}v_1 \\ T(v_2) &= a_{12}v_1 + a_{22}v_2 \\ &\vdots \\ T(v_n) &= a_{1n}v_1 + \cdots + a_{nn}v_n \end{aligned}$$

   Recall for an upper triangular matrix the eigenvalues are along the diagonal so we want to show $T(v_j) = a_{jj}v_j$ for each j. By the above formulas note this is equivalent to showing $a_{ij} = 0$ for all $i \le j - 1$. We do this inductively. This is already true for $j = 1$. Now assume $T(v_i) = a_{ii}v_i$ for $i = 1, \ldots, j - 1$. Since T is normal we also have $T^*(v_i) = \overline{a_{ii}}v_i$. Then $a_{ij} = \langle T(v_j), v_i\rangle = \langle v_j, T^*(v_i)\rangle = \langle v_j, \overline{a_{ii}}v_i\rangle = 0$ by orthonormality for all $i \le j - 1$. Thus $T(v_j) = a_{jj}v_j$ and we have closed the induction. This implies $[T]_{\mathcal{B}}^{\mathcal{B}}$ is diagonal, as desired.

$\square$

Lecture 13 April 27, Even More Unitary/Orthogonal Operators

Remark 13.1. Note that by characterizations of orthogonal/unitary matrices, if $A = QDQ^{-1}$ for diagonal D and Q orthogonal/unitary, then the columns of Q are precisely the eigenvectors.

Note the eigenvalues also satisfy $\lambda = \langle Av, v\rangle$ for some v, since if v is an eigenvector, we have $\langle Av, v\rangle = \lambda\|v\|^2$, so by normalizing, we get the desired relation.

The goal today is to try to understand how to classify orthogonal matrices by decomposing them. This is relatively simple to do for $n = 1, 2$.

Example 13.2 (Classifying Small Orthogonal Matrices)
We have for

- $n = 1$: Since we need $A^{-1} = A^T$, we can see the only possibilities are $\begin{pmatrix}1\end{pmatrix}$ or $\begin{pmatrix}-1\end{pmatrix}$. Note we can think of this as corresponding to a rotation (which in 1D does nothing, since there is no axis to rotate about) or a reflection, which just flips the point about the origin.
- $n = 2$: Here we can classify by the orthogonal matrices by considering an arbitrary 2x2 matrix and solve the system of equations corresponding to $A^TA = I$:

$$\text{Let } A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \text{ then } A^TA = I \implies \begin{pmatrix} a^2+c^2 & ab+cd \\ ab+cd & b^2+d^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Then we can let $a = \cos\theta$, $c = \sin\theta$ and $b = \cos\phi$, $d = \sin\phi$ for some values of θ and ϕ, then $ab+cd = 0$ becomes $\cos\theta\cos\phi + \sin\theta\sin\phi = \cos(\theta-\phi) = 0$. This gives two solutions $\theta - \phi = \pm\pi/2$, which you can check gives us the two cases

$$A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \quad \text{or} \quad A = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$$

The first case corresponds to a rotation of angle θ, and you can check the second case is a reflection, and is actually orthogonally equivalent (via diagonalization) to the matrix

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Then here there are two eigenspaces corresponding to $\lambda = \pm 1$, which we denote E_1 and E_{-1}. Then $\mathbb{R}^2 = E_1 \oplus E_{-1}$. Note these correspond to the 1D "rotations" and "reflections".

It turns out this is actually enough to decompose orthogonal matrices fully, but we won't be able to see the full explanation yet.

Remark 13.3. Let $A \in M_{n\times n}(\mathbb{R})$. Consider the characteristic polynomial of A, which is $f_A(t)$. This polynomial clearly splits over $\mathbb{C}$ by the Fundamental Theorem of Algebra, so it has n real and complex roots, up to multiplicity. Real roots of the polynomial gives us an eigenvalue, corresponding to a dimension 1 eigenspace.

On the other hand, consider a complex root $\lambda = a + bi$. Then when considering the matrix as a complex matrix, it is a complex eigenvalue, so there exists nonzero $w \in \mathbb{C}^n$ such that $Aw = \lambda w$. Let $w = v_1 + iv_2$ (which we can do since we are explicitly working in $\mathbb{C}^n$), so $A(v_1 + iv_2) = (a + bi)(v_1 + iv_2) = (av_1 - bv_2) + i(av_2 + bv_1)$. Thus $Av_1 = av_1 - bv_2$ and $Av_2 = av_1 + bv_1$. Hence $\text{span}(v_1, v_2)$ is an invariant subspace.

Thus given such a matrix we can essentially keep on extracting dimension 1 or 2 invariant subspaces. We follow a very similar argument to the proof of Schur decomposition (specifically the version in Theorem 10.9), where we take a real or complex root of the characteristic polynomial, find the corresponding invariant space, and quotient out by it. We are left with a quotient space, and an operator restricted to the quotient, whose characteristic polynomial divides the original characteristic polynomial, and we repeat the process. As in the proof of Theorem 10.9, we can see this gives us a block triangular matrix:

$$\begin{pmatrix} \boxed{\lambda_1} & & & & & * \\ & \ddots & & & & \\ & & \boxed{\lambda_k} & & & \\ & & & \boxed{A_1} & & \\ & & & & \ddots & \\ 0 & & & & & \boxed{A_m} \end{pmatrix}$$

Remark 13.4. It turns out for orthogonal matrices these blocks will exactly be the 1×1 and 2×2 orthogonal matrices, and furthermore, it is not only block upper triangular, but block diagonal (which we can't guarantee here since we had to take quotients). I haven't really figured out why myself, so for now I'll leave it like this.

Lecture 14 April 29

Review.

Lecture 15 May 1

Midterm.

Lecture 16 May 4, Orthogonal Projections

Recall for any subspace W, we can find an direct complement W' (not necessarily unique). This motivates the following definition:

Definition 16.1 (Projection) — Let V be a vector space. Let W, W' be subspaces such that $V = W \oplus W'$. Then for every $v \in V$ we can uniquely write $v = w + w'$ where $w \in W$ and $w' \in W'$. We define the **projection** of V onto the subspace W to be the linear operator $T : V \to V$ defined by $T(v) = T(w + w') = w$.

Theorem 16.2 (Characterizations of Projections)
Let V be a vector space and $T : V \to V$ be a linear operator. The following are equivalent:

(i) T is a projection.

(ii) $T^2 = T$

Proof. (i) $\implies$ (ii) Suppose T is a projection onto some subspace W with some direct complement W'. Then for each $v \in V$ we can write $v = w+w'$ for $w \in W$ and $w' \in W'$. Then $T^2(v) = T(T(v)) = T(w) = w = T(v)$, so $T^2 = T$.

(ii) $\implies$ (i) Now suppose $T^2 = T$. Suppose we show $V = \operatorname{im}(T) \oplus \ker(T)$. Then for any $v \in V$ we can write $v = w + w'$ where $w \in \operatorname{im}(T)$ and $w' \in \ker(T)$, so $T(v) = T(w)$. There also exists some $u \in V$ such that $T(u) = w$. But since $T^2 = T$ we have $T(v) = T(w) = T(T(u)) = T(u) = w$, so T is a projection.

It remains to show $V = \operatorname{im}(T) \oplus \ker(T)$. Note that given $v \in V$, we have $T(v) \in \operatorname{im}(T)$ be definition and $T(v - T(v)) = T(v) - T^2(v) = 0$, so $v - T(v) \in \ker(T)$. Hence $v \in \operatorname{im}(T) + \ker(T)$, so $\operatorname{im}(T) + \ker(T) = V$. Now if $v \in \operatorname{im}(T) \cap \ker(T)$, there exists $u \in V$ such that $v = T(u)$, but also $T(v) = 0$, so $0 = T^2(u) = T(u) = v$, and hence $\operatorname{im}(T) \cap \ker(T) = \{0\}$, which concludes the proof. $\square$

Remark 16.3. Note that because the direct complement is not unique, the projection onto a subspace is also not unique. This does not affect how the map acts on the subspace it's projecting onto but will affect how it acts on the rest of the space. For example, consider projecting $\mathbb{R}^2$ onto a line through the origin. Then any other line through the origin which is not parallel to the first line is a direct complement, so depending on which line you choose, the projection map is 0 on a different line.

It is also easy to show that if T is a projection then $V = \operatorname{im}(T) \oplus \ker(T)$ but the converse is not true, as maps which project onto a subspace but then scale the output also satisfy this.

We saw that the non-uniqueness of direct complements makes it sometimes easier to work with the quotient space, but if we have an inner product space, we can use the orthogonal complement, which is unique:

Definition 16.4 (Orthogonal Projection) — Let $(V, \langle \cdot, \cdot \rangle)$ be a finite dimensional inner product space and $T : V \to V$ be a projection. We say T is an **orthogonal projection** if $\ker(T) = \operatorname{im}(T)^\perp$.

Remark 16.5. Since V is finite dimensional we also get for free that $\operatorname{im}(T) = \ker(T)^\perp$. Note that it makes sense (in finite dimensions) to say the "the orthogonal projection onto a subspace W" without any additional information since we can use the decomposition $V = W \oplus W^\perp$ and the uniqueness of the orthogonal complement guarantees this is the only orthogonal projection onto W.

The name is also misleading since orthogonal projections are usually not orthogonal maps (if they project onto any proper subspace they are not even invertible).

Theorem 16.6 (Characterizations of Orthogonal Projections)
Let $(V, \langle \cdot, \cdot \rangle)$ be a finite dimensional inner product space and $T : V \to V$ be a linear operator. Then the following are equivalent:

(i) T is an orthogonal projection.

(ii) T has an adjoint and $T^2 = T = T^*$. (i.e. T is a self-adjoint projection).

(iii) For all $w \in \operatorname{im}(T)$, we have $\|T(v) - v\| \leq \|w - v\|$. (i.e. the orthogonal projection projects onto the closest vector in the subspace.)

Remark 16.7. These equivalences hold in infinite dimensional spaces as well when the space is a Hilbert space (complete inner product space). It turns out that the minimization property of (iii) is actually the more natural way to define orthogonal projections in infinite dimensions as you can use this to define projections for some more general sets which are not subspaces, and you recover an orthogonal projection when the set is a subspace.

Proof. (i) $\implies$ (ii) Since T is a projection we automatically have $T^2 = T$. We also have $V = \operatorname{im}(T) \oplus \ker(T)$. Since it's an orthogonal projection we also have $\ker(T) = \operatorname{im}(T)^\perp$ and $\operatorname{im}(T) = \ker(T)^\perp$. Now given $x, y \in V$, we can write the decompositions $x = x_1 + x_2$ and $y = y_1 + y_2$, where $x_1, y_1 \in \operatorname{im}(T)$ and $x_2, y_2 \in \ker(T)$. Then by orthogonality we have

$$\langle T(x), y \rangle = \langle x_1, y \rangle = \langle x_1, y_1 \rangle + \langle x_1, y_2 \rangle = \langle x_1, y_1 \rangle$$

$$\langle x, T(y) \rangle = \langle x, y_1 \rangle = \langle x_1, y_1 \rangle + \langle x_2, y_1 \rangle = \langle x_1, y_1 \rangle$$

Hence $\langle T(x), y \rangle = \langle x, T(y) \rangle$, so T has an adjoint, namely itself.

(ii) $\implies$ (i) This follows from the following facts about the adjoint in finite dimensional spaces (and the fact that we assume T is self-adjoint here):

$$\ker(T^*) = \operatorname{im}(T)^\perp \quad \text{and} \quad \operatorname{im}(T^*) = \ker(T)^\perp$$

See 115AH HW10 for details.

(i) $\implies$ (iii) Let $w \in \text{im}(T)$. We have $w - T(v) \in \text{im}(T)$ and $T(v) - v \in \ker(T)$ so they are orthogonal. Then by the Pythagorean theorem we have

$$\|w - v\|^2 = \|w - T(v)\|^2 + \|T(v) - v\|^2 \geq \|T(v) - v\|^2$$

and taking a square root gives the desired inequality.

(iii) $\implies$ (i) Let $w \in \text{im}(T)$. Since V is finite dimensional we have $V = \text{im}(T) \oplus \text{im}(T)^\perp$. Then for any $v \in V$ we can write $v = u + u'$ for $u \in \text{im}(T)$ and $u' \in \text{im}(T)^\perp$. Again we have

$$\begin{aligned}
\|T(v) - v\|^2 &\leq \|w - v\|^2 \\
\|(T(v) - u) - u'\|^2 &\leq \|(w - u) - u'\|^2 \\
\|T(v) - u\|^2 + \|u'\|^2 &\leq \|w - u\|^2 + \|u'\|^2 \\
\|T(v) - u\|^2 &\leq \|w - u\|^2
\end{aligned}$$

Since this is true for any choice of w, pick $w = u$, so we have $\|T(v) - u\|^2 \leq 0$, which implies $T(v) - u = 0$, and hence T is a projection onto $\text{im}(T)$. But then by definition of the projection and the decomposition we must have $\ker(T) = \text{im}(T)^\perp$. Hence we have that T is an orthogonal projection. $\square$

Lecture 17 May 6, Spectral Theorem Part II

The goal now is to use orthogonal projections to decompose diagonalizable operators based on their eigenspaces. Since these notes already refer to the result about diagonalizability of operators over inner product spaces as the "Spectral Theorem," we call this part II.

Theorem 17.1 (Spectral Theorem, Part II)
Let $(V, \langle \cdot, \cdot \rangle)$ be a finite dimensional inner product space and let $T : V \to V$ be a linear operator. If $F = \mathbb{C}$ let T be normal and if $F = \mathbb{R}$ let T be self-adjoint. Let $\lambda_1, \ldots, \lambda_k$ be the distinct eigenvalues of T and let $W_1, \ldots, W_k$ be the corresponding eigenspaces. Let T_i denote the orthogonal projection onto W_i for each i. Then we have:

(i) $V = W_1 \oplus + \cdots + \oplus W_k$

(ii) For each i we have $W_i^\perp = \bigoplus_{j \neq i} W_j$

(iii) $T_i T_j = \delta_{ij} T_i = \begin{cases} T_i & j = i \\ 0 & j \neq i \end{cases}$ for all i, j.

(iv) $\mathbf{1}_V = T_1 + \cdots + T_k$

(v) $T = \lambda_1 T_1 + \cdots + \lambda_k T_k$.

Remark 17.2. Note that when $F = \mathbb{C}$ and T is self-adjoint, this additionally implies the eigenvalues are real. However, I see no reason to separate this into three cases as Professor Gao stated in lecture.

Proof. This whole theorem was proved at the end of 115AH last quarter.

(i) By the Spectral Theorem, Part I, V has an orthonormal eigenbasis, and by organizing the basis vectors by the eigenspace, this statement follows (we showed this in 115AH).

(ii) Recall that eigenvectors corresponding to different eigenvalues are orthogonal. Then we have $W_i \perp W_j$ whenever $i \neq j$ so it follows $\bigoplus_{j \neq i} W_j \subseteq W_i^\perp$. Now given $v \in W_i^\perp$, we can write $v = w_1 + \cdots + w_k$ where $w_i \in W_i$, and $0 = \langle v, w_i \rangle = \langle w_1, w_i \rangle + \cdots + \langle w_k, w_i \rangle = \langle w_i, w_i \rangle$. Hence $w_i = 0$, so $v \in \bigoplus_{j \neq i} W_j$, and we are done.

(iii) By (i) and (ii), when we write $v = w_1 + \cdots + w_k$, we have $T_i(v) = w_i$. This then follows by computation.

(iv)-(v) These are just computation.

□

We also state the matrix version:

Theorem 17.3

Let $A \in M_{n\times n}(F)$. If $F = \mathbb{C}$ let A be normal and if $F = \mathbb{R}$ let A be symmetric. Let $\lambda_1, \ldots, \lambda_k$ be the distinct eigenvalues of A. Let $n_1, \ldots, n_k$ be the sizes of the corresponding eigenspaces. Then there exists $Q \in M_{n\times n}(F)$ such that

(i) $A = Q \begin{pmatrix} \lambda_1 I_{n_1} & & \\ & \ddots & \\ & & \lambda_k I_{n_k} \end{pmatrix} Q^{-1}$ where Q is unitary if $F = \mathbb{C}$ and orthogonal if $F = \mathbb{R}$.

(ii) $\begin{pmatrix} \lambda_1 I_{n_1} & & \\ & \ddots & \\ & & \lambda_k I_{n_k} \end{pmatrix} = \lambda_1 \begin{pmatrix} I_{n_1} & \\ & \end{pmatrix} + \cdots + \lambda_k \begin{pmatrix} & \\ & I_{n_k} \end{pmatrix}$

Remark 17.4. Similarly if, $F = \mathbb{C}$ and A is Hermitian, this only additionally gives us that the eigenvalues are real. Professor Gao claims this makes the matrix orthogonally diagonalizable $Q^{-1} = Q^T$ but this is **incorrect**.

Lecture 18 May 8, Consequences of the Spectral Theorem

Let $(V, \langle \cdot, \cdot \rangle)$ be a finite dimensional inner product space and let $T : V \to V$ be a linear operator.

Proposition 18.1

If $F = \mathbb{C}$, then T is normal iff $T^* = g(T)$ for some polynomial g.

Proof. ($\Longrightarrow$) Since T is normal we have $T = \lambda_1 T_1 + \cdots + \lambda_k T_k$. Since each T_i is self-adjoint (it is an orthogonal projection) we have $T^* = \overline{\lambda}_1 T_1 + \cdots + \overline{\lambda}_k T_k$. Consider the Lagrange polynomials defined by:

$$f_i(x) := \prod_{\substack{0 \le k \le n \\ k \neq i}} \frac{x - \lambda_k}{\lambda_i - \lambda_k}$$

which satisfy

$$f_i(\lambda_j) = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

Then define

$$g(x) = \sum_{i=1}^{k} \overline{\lambda}_i f_i(x)$$

so $g(\lambda_i) = \overline{\lambda}_i$. (This is known as **Lagrange interpolation**.) Now by the homework (Section 6.6 Exercise 7.(a)) we have

$$g(T) = \sum_{i=1}^{k} g(\lambda_i) T_i = \sum_{i=1}^{k} \overline{\lambda}_i T_i = T^*$$

($\Longleftarrow$) Suppose there is some polynomial g such that $T^* = g(T)$. Note that power of T commute with each other, and hence by linearity, polynomials of T commute with T. Hence $TT^* = Tg(T) = g(T)T = T^*T$. $\square$

Proposition 18.2

If $F = \mathbb{C}$, then T is unitary iff T is normal and $|\lambda_i| = 1$ for each eigenvalue of T.

Proof. ($\Longrightarrow$) If T is unitary, it is normal, since $TT^{-1} = T^{-1}T = \mathbf{1}_V$ and $T^{-1} = T^*$. Since T is normal, if λ is an eigenvalue of T and v is a corresponding eigenvector, then $T^*(v) = \overline{\lambda} v$. Thus we have $TT^*(v) = T(\overline{\lambda} v) = \lambda \overline{\lambda} v = |\lambda|^2 v$. But $TT^*(v) = \mathbf{1}_V(v) = v$, so we must have $|\lambda|^2 = 1$.

($\Longleftarrow$) Suppose T is normal and $|\lambda_i| = 1$ for each eigenvalue of T. Then we have $T = \lambda_1 T_1 + \cdots + \lambda_k T_k$ and $T^* = \overline{\lambda}_1 T_1 + \cdots + \overline{\lambda}_k T_k$, so using that $T_i T_j = \delta_{ij} T_i$, we have

$$TT^* = (\lambda_1 T_1 + \cdots + \lambda_k T_k)(\overline{\lambda}_1 T_1 + \cdots + \overline{\lambda}_k T_k) = |\lambda_1|^2 T_1 + \cdots + |\lambda_k|^2 T_k = T_1 + \cdots + T_k = \mathbf{1}_V$$

We can do a similar calculation for T^*T (which recall we do not get for free in infinite dimensions). $\square$

Theorem 18.3

Suppose T, U are normal operators on V. Then $TU = UT$ iff they are simultaneously orthogonally diagonalizable (i.e. there exists an orthonormal basis which is an eigenbasis for both T and U.)

Proof. ($\implies$) Since T is normal, by the Spectral Theorem Part I, if the distinct eigenvalues of T are $\lambda_1, \ldots \lambda_k$, then $V = E_{\lambda_1}(T) \oplus \cdots \oplus E_{\lambda_k}(T)$, where $E_{\lambda_i}(T)$ is the eigenspace corresponding to λ_i of T. We want to show each $E_{\lambda_i}(T)$ is U invariant as well.

Given $v \in E_{\lambda_i}(T)$, we have $T(v) = \lambda_i v$, so $T(U(v)) = U(T(v)) = U(\lambda_i v) = \lambda_i U(v)$, and hence $U(v) \in E_{\lambda_i}(T)$, so $E_{\lambda_i}(T)$ is U-invariant. Then $U|_{E_{\lambda_i}(T)}$ is well-defined, and since U is normal, so must $U|_{E_{\lambda_i}(T)}$. Hence this restrict operator is orthogonally diagonalizable, and let $\mathcal{B}_i$ be an eigenbasis of $E_{\lambda_i}(T)$ for U. Note that by definition each of these basis vectors are automatically eigenvectors of T. By taking a union of all these bases, by the direct sum, we get a basis for V, which is both an eigenbasis of U and T, and we are done.

($\impliedby$) Suppose there exists an orthonormal basis $\mathcal{B} = \{v_1, \ldots v_n\}$ which are eigenbases for both T and U. For each i let λ_i be the eigenvalue for T and let μ_i be the eigenvalue for U. Then for each i we have $T(U(v_i)) = \lambda_i \mu_i v_i = \mu_i \lambda_i v_i = U(T(v_i))$. Then by linearity it follows it holds for every vector in V, so $TU = UT$. $\square$

Remark 18.4. We omit the matrix version of the proof that Gao attempted to give in lecture because it doesn't really avoid any hurdles and is more opaque.

Lecture 19 May 11, Bilinear and Quadratic Forms

Let V be a vector space over F. In general F can be taken to be any field, but most of the results about diagonalization and decomposition break when in a field of characteristic 2. As before, whenever we assume the space is an inner product space, we then restrict ourselves to $F = \mathbb{R}$ or $\mathbb{C}$.

Quadratic forms are in a naïve sense polynomials in n variables where each term is degree 2. They appear in so many different contexts that there isn't really one good way to intuitively think of them. We will see that the "correct" way to work with them is through the lens of bilinear forms.

For most of the quadratic forms stuff, I will be using a combination of Axler's Linear Algebra Done Right and Treil's Linear Algebra Done Wrong.

Definition 19.1 (Bilinear Form) — A **bilinear form** on V is a function $B : V \times V \to F$ which is linear in each argument, i.e. $v \mapsto B(v, w)$ and $w \mapsto B(v, w)$ are both linear. The set of bilinear forms on V is a vector space, which we denote by $\mathrm{Bil}(V)$.

Definition 19.2 (Matrix of a Bilinear Form) — Suppose V is finite dimensional and let $\mathcal{B} = \{v_1, \ldots, v_n\}$ be a basis for V. Let $B : V \times V \to F$ be a bilinear form on V. Let $B_{ij} = B(v_i, v_j)$. Then we denote the matrix of B with respect to $\mathcal{B}$ to be

$$[B]_{\mathcal{B}} = \begin{pmatrix} B_{11} & \cdots & B_{1n} \\ \vdots & \ddots & \vdots \\ B_{n1} & \cdots & B_{nn} \end{pmatrix}$$

Proposition 19.3

Let B be as above. Now given $v, w \in V$ let $[v]_{\mathcal{B}}, [w]_{\mathcal{B}} \in F^n$ denote the coordinates in the $\mathcal{B}$ basis. Then we have

$$B(v, w) = ([v]_{\mathcal{B}})^T [B]_{\mathcal{B}} [w]_{\mathcal{B}}$$

Essentially, in matrix form, the bilinear form takes in vectors by multiplying them in on both sides.

Proposition 19.4

If $\dim(V) < \infty$, $\dim(\mathrm{Bil}(V)) = \dim(V)^2$.

Proposition 19.5

Suppose V is finite dimensional and $\mathcal{B}$ be a basis. Let $B : V \times V \to F$ be a bilinear form and $T : V \to V$ be a linear operator. Define the bilinear forms A and C by $A(v, w) = B(T(v), w)$ and $C(v, w) = B(v, T(w))$. Then we have

$$[A]_{\mathcal{B}} = ([T]_{\mathcal{B}})^T [B]_{\mathcal{B}} \quad \text{and} \quad [C]_{\mathcal{B}} = [B]_{\mathcal{B}} [T]_{\mathcal{B}}$$

Proposition 19.6 (Change of Basis)
Suppose V is finite dimensional and $\mathcal{B}, \mathcal{C}$ are bases. Let B be a bilinear form on V. Then

$$[B]_{\mathcal{C}} = ([\mathbf{1}_V]_{\mathcal{C}}^{\mathcal{B}})^T [B]_{\mathcal{B}} [\mathbf{1}_V]_{\mathcal{C}}^{\mathcal{B}}$$

We omit the proofs of these facts as they are just computation with coordinates and index management. See Axler for the details.

We then define quadratic forms in terms of bilinear forms:

Definition 19.7 (Quadratic Form) — We say $q : V \to F$ is a **quadratic form** if there exists a bilinear form $B : V \times V \to F$ such that $q(v) = B(v, v)$.

We will most commonly look at quadratic forms on F^n.

Proposition 19.8 (Quadratic Forms on F^n)
Let $q : F^n \to F$. Then q is a quadratic form iff there exists $A_{ij} \in F$ for all i, j such that

$$q(x_1, \ldots, x_n) = \sum_{i=1}^{n} \sum_{j=1}^{n} A_{ij} x_i x_j$$

In other words, for $x \in F^n$, there exists some matrix $A \in M_{n \times n}(F)$ such that

$$q(x) = x^T A x$$

Proof. ($\Longrightarrow$) Let q be a quadratic form, so $q(v) = B(v, v)$ for some bilinear form B. Then define $A_{ij} := B(e_i, e_j)$. Then by linearity in each term we get the desired expansion.

($\Longleftarrow$) Define the following function which you can check is a bilinear form: For $v = (x_1, \ldots, x_n), w = (y_1, \ldots, y_n) \in F^n$, let

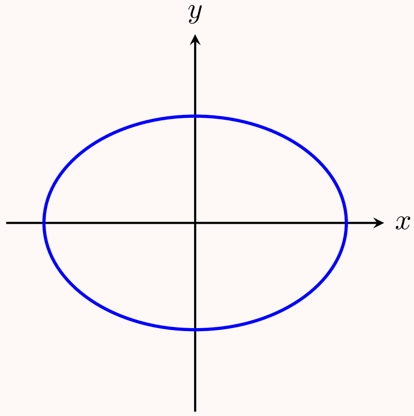
$$B(v, w) = \sum_{i=1}^{n} \sum_{j=1}^{n} A_{ij} x_i y_j$$

Then clearly $q(v) = B(v, v)$. $\square$

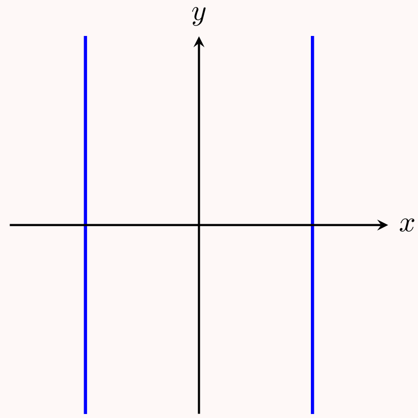
Example 19.9

Consider $V = \mathbb{R}^2$ and quadratic forms $q : \mathbb{R}^2 \to \mathbb{R}$. Here they take the form $q(x_1, x_2) = ax_1^2 + bx_1x_2 + cx_2^2$. One way to visualize quadratic forms is through their level sets. It turns out there are three primary types of quadratic forms, and the rest are stretched/rotated versions:

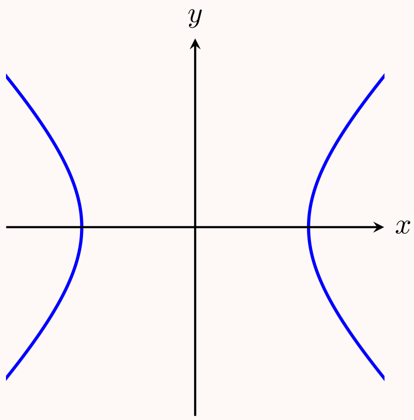
- Ellipses: $q(x_1, x_2) = ax_1^2 + bx_2^2$



- Parallel lines: $q(x_1, x_2) = x_1^2$



- Hyperbolas: $q(x_1, x_2) = x_1^2 - x_2^2$



You can try graphing a few different quadratic forms to see that the ones without the cross terms are the ones that are "aligned" with the x-y axes, and otherwise they are aligned along some rotated axes. Ideally we'd like to know how to describe the axes, which we will see ends up being a problem of "diagonalizing" the quadratic form.

Before we do that, let us show that there is in fact a bijection between symmetric bilinear forms and quadratic forms. By definition every bilinear form defines a quadratic form and every quadratic form comes from a bilinear form, but we need to show that that there is a unique way to do this with symmetric bilinear forms.

Definition 19.10 (Symmetric Bilinear Form) — We say a bilinear form B is symmetric if $B(v, w) = B(w, v)$ for all $v, w \in V$. The set of symmetric bilinear forms on V is a subspace, which we denote by $\mathrm{Sym}^2(V)$.

Definition 19.11 (Diagonalization of Bilinear Forms) — We say a bilinear form B is diagonalizable if there exists some basis $\mathcal{B}$ such that $[B]_{\mathcal{B}}$ is a diagonal matrix.

Theorem 19.12 (Diagonalizing Symmetric Bilinear Forms)
Let V be finite dimensional and $B \in \mathrm{Bil}(V)$. The following are equivalent:

(i) B is a symmetric bilinear form on V.

(ii) $[B]_{\mathcal{B}}$ is a symmetric matrix for every basis $\mathcal{B}$.

(iii) $[B]_{\mathcal{B}}$ is a symmetric matrix for some basis $\mathcal{B}$.

(iv) $[B]_{\mathcal{B}}$ is a diagonal matrix for some basis $\mathcal{B}$.

If additionally $F = \mathbb{R}$ and V is an inner product space, then $[B]_{\mathcal{B}}$ is a diagonal matrix for some orthonormal basis $\mathcal{B}$.

Proof. (i) $\Longrightarrow$ (ii) Let $\mathcal{B} = \{v_1, \ldots, v_n\}$ be any basis. Then $B(v_i, v_j) = B(v_j, v_i)$ for all i, j. But these are the i, jth entries of the matrix of B, and hence the matrix is symmetric.

(ii) $\Longrightarrow$ (iii) Trivial.

(iii) $\Longrightarrow$ (i) Let $\mathcal{B} = \{u_1, \ldots, u_n\}$ be a basis for which the matrix of B is symmetric. Then for any $v, w \in V$ we can write $v = a_1u_1 + \cdots + a_nu_n$ and $w = b_1u_1 + \cdots + b_nu_n$ for $a_i, b_j \in F$, so

$$\begin{aligned} B(v, w) &= \sum_{i=1}^{n} \sum_{j=1}^{n} a_i b_j B(u_i, u_j) \\ &= \sum_{j=1}^{n} \sum_{i=1}^{n} b_j a_i B(u_j, u_i) \\ &= B(w, v) \end{aligned}$$

(i) $\Longrightarrow$ (iv) We induct on the size of the matrix. This is true for $n = 1$ since every 1×1 matrix is diagonal. Now assume the implication holds for $n - 1$. Let B be a symmetric bilinear form on V where $\dim V = n$. If $B = 0$ then every matrix representation is the zero matrix, so we are done, so assume $B \neq 0$. Then there exists $v, w \in V$ such that $B(v, w) \neq 0$. By using bilinearity and symmetry, we get

$$2B(v, w) = B(v + w, v + w) - B(v, v) - B(w, w)$$

Then the RHS cannot be zero, so there exists at least one nonzero term on the RHS, and thus there exists $u \in V$ such that $B(u, u) \neq 0$. Now let

$$W = \{w \in V : B(w, u) = 0\}$$

Then W is the kernel of the linear functional $w \mapsto B(w, u)$. But since $u \notin W$, this linear functional is not identically 0, and by Rank-Nullity $\dim(W) = n - 1$. By restricting ourselves to this subspace, we get the bilinear form $B|_{W \times W}$, and by the induction hypothesis there exists some basis $\{v_1, \ldots, v_{n-1}\}$ such that the matrix of $B|_{W \times W}$ is diagonal.

Now since $u \notin W$, $\{v_1, \ldots, v_{n-1}, u\}$ is a basis for V, and by the definition of W we have $B(v_i, u) = 0$ for each i. By symmetry, it follows that the matrix of B with respect to this basis is diagonal.

(iv) $\implies$ (iii) Trivial.

If V is a real inner product space, use the Spectral Theorem for Matrices to conclude the fact immediately. $\square$

Remark 19.13. Recall that the change of basis formula for bilinear forms is different than the change of basis formula for linear operators, and hence the meaning of diagonalization is different as well, except when we have a real inner product space, in which we can diagonalize by an orthogonal Q (so $Q^T = Q^{-1}$).

Definition 19.14 (Alternating Bilinear Form) — We say a bilinear form B is alternating if $B(v, w) = -B(w, v)$ for all $v, w \in V$. The set of alternating bilinear forms on V is a subspace, which we denote by $\mathrm{Alt}^2(V)$.

Proposition 19.15

B is an alternating bilinear form iff $B(v, v) = 0$ for all $v \in V$.

Proof. ($\implies$) Trivial.

($\impliedby$) For $v, w \in V$ we have $0 = B(v + w, v + w) = B(v, v) + B(v, w) + B(w, v) + B(w, w) = B(v, w) + B(w, v)$, as desired. $\square$

Proposition 19.16

$\mathrm{Bil}(V) = \mathrm{Sym}^2(V) \oplus \mathrm{Alt}^2(V)$

Proof. Given $B \in \mathrm{Bil}(V)$, define

$$A(v, w) = \frac{B(v, w) + B(w, v)}{2} \in \mathrm{Sym}^2(V), \quad C(v, w) = \frac{B(v, w) - B(w, v)}{2} \in \mathrm{Alt}^2(V)$$

Thus $\mathrm{Bil}(V) = \mathrm{Sym}^2(V) + \mathrm{Alt}^2(V)$. Let $B \in \mathrm{Sym}^2(V) \cap \mathrm{Alt}^2(V)$. Then $B(v, w) = B(w, v) = -B(w, v)$. Hence $B(w, v) = 0$ for all $w, v \in V$, and thus $B = 0$. $\square$

Theorem 19.17 (Unique Correspondence between Symmetric Bilinear and Quadratic Forms)

For every quadratic form $q : V \to F$ there exists a unique symmetric bilinear form $B : V \times V \to F$ such that $q(v) = B(v, v)$. On the other hand every symmetric bilinear form defines a quadratic form in this manner.

Proof. The second statement is true by definition. Let $q : V \to F$ be a quadratic form. Then there exists some bilinear form B such that $q(v) = B(v, v)$. By the decomposition there exists $A \in \mathrm{Sym}^2(V)$ and $C \in \mathrm{Alt}^2(V)$ such that $B = A + C$, so $q(v) = A(v, v) + C(v, v) = A(v, v)$. Now suppose there is another $A' \in \mathrm{Sym}^2(V)$ such that $q(v) = A'(v, v)$. Then $(A - A')(v, v) = 0$ for all $v \in V$ so $A - A' \in \mathrm{Alt}^2(V)$, but by subspace properties $A - A' \in \mathrm{Sym}^2(V)$ as well, so $A - A' \in \mathrm{Sym}^2(V) \cap \mathrm{Alt}^2(V) = \{0\}$, so $A = A'$, and we have shown uniqueness. $\square$

Lecture 20 May 13, Diagonalizing Quadratic Forms

Let V be a vector space over F. The same remarks about the underlying field from last time still apply here.

Last time we showed you can always diagonalize a symmetric bilinear form (on a finite dimensional space), and if the underlying vector space is an inner product space and $F = \mathbb{R}$, then you can orthogonally diagonalize it. Similarly we can do this for quadratic forms.

Theorem 20.1 (Diagonalizing Quadratic Forms)
Let V be finite dimensional and $q : V \to F$ be a quadratic form. Then there exists a basis $\mathcal{B} = \{v_1, \ldots, v_n\}$ and $\lambda_1, \ldots, \lambda_n \in F$ such that $q(x_1v_1 + \cdots + x_nv_n) = \lambda_1 x_1^2 + \cdots + \lambda_n x_n^2$ for all $x_i \in F$.

If additionally $F = \mathbb{R}$ and V is an inner product space, then the basis can be chosen to be orthonormal.

Proof. By Theorem 19.17 there exists a unique symmetric bilinear form B such that $q(v) = B(v, v)$. Now by Theorem 19.12 there exists some basis $\mathcal{B}$ such that the matrix of B is diagonal. Then

$$\begin{aligned} q(x_1v_1 + \cdots + x_nv_n) &= B(x_1v_1 + \cdots + x_nv_n, x_1v_1 + \cdots + x_nv_n) \\ &= \sum_{i=1}^{n} \sum_{j=1}^{n} x_i x_j B(v_i, v_j) \\ &= \sum_{i=1}^{n} x_i^2 B(v_i, v_i) \end{aligned}$$

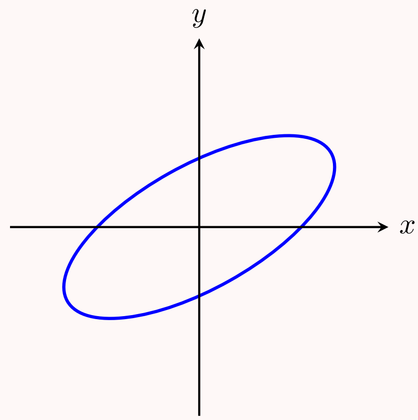
And if we let $\lambda_i := B(v_i, v_i)$, we get the desired result. Again by the same theorem, if $F = \mathbb{R}$ and V is an inner product space, then we can take the basis to be orthonormal. $\square$

We now return to looking at quadratic forms on F^n and attempting to diagonalize them.

Example 20.2

Due to the fact that quadratic forms are degree 2 homogeneous polynomials (a polynomial such that every term has degree 2), if the dimension is low enough we can actually diagonalize them naïvely by completing squares.

Consider $V = \mathbb{R}^2$ and $q(x_1, x_2) = 7x_1^2 - 10x_1x_2 + 4x_2^2$. By completing the square, we get $q(x_1, x_2) = 7(x_1 - \frac{5}{7}x_2)^2 + \frac{3}{7}x_2^2$, so by taking the change of basis $y_1 = x_1 - \frac{5}{7}x_2$ and $y_2 = x_2$, it becomes clear that q defines a rotated ellipse, and indeed:



However note that this new basis is not orthogonal, and it should be visually obvious that the "axes" the ellipse is aligned with does not have slope 0 and 5/7. So while this method can sometimes take less computation, it gives us less geometric information.

Example 20.3

On the other hand, if we orthogonally diagonalize the quadratic form, we can just diagonalize the matrix normally, and additionally having an orthonormal basis means, it tells us exactly how the ellipsoid/hyperboloid is stretched, and along which axes it is aligned, as the change of basis is a rigid transformation preserving lengths and angles.

For example, let $q : \mathbb{R}^3 \to \mathbb{R}$ be defined by $q(x_1, x_2, x_3) = 2x_1^2+2x_2^2+2x_3^2+2x_1x_2+2x_2x_3+2x_3x_1$. We can then write q in terms of the matrix of a symmetric bilinear form in the standard basis as follows:

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \quad q(x) = x^T Ax, \quad x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

The general strategy for this is to write down all the diagonal terms first, and then split the cross terms in half and write each half on one side of the diagonal. Then it is an easy exercise to orthogonally diagonalize the matrix:

$$A = Q^T DQ, \quad D = \begin{pmatrix} 4 & & \\ & 1 & \\ & & 1 \end{pmatrix}, \quad Q = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & -\frac{2}{\sqrt{6}} \end{pmatrix}$$

Lecture 21 May 15, Sylvester's Law of Inertia and Criterion

We saw with linear operators, the entries in any diagonal matrix representation is invariant (up to permutation) under change of bases. Unfortunately, this does not hold for bilinear forms as the change of basis formula is different. However, we do get an invariant in terms of the signs of the diagonal entries.

Let V be a vector space over F.

Theorem 21.1 (Sylvester's Law of Inertia)
Let V be finite dimensional, $F = \mathbb{R}$, and $B : V \times V \to \mathbb{R}$ be a symmetric bilinear form. Then the number of positive, negative, and zero entries on the diagonal of a diagonal matrix representation of B is respectively invariant under the choice of basis. That is, given two bases for which the matrix of B is diagonal, they have the same number of positive, negative and zero entries on the diagonal, respectively.

Proof. Let $\mathcal{B} = \{v_1, \ldots, v_n\}$ and $\mathcal{C} = \{w_1, \ldots, w_n\}$ be two bases for which the matrix of B is diagonal. Let n_+, n_-, n_0 respectively denote the number of positive, negative, and zero entries along the diagonal of the matrix of B with respect to $\mathcal{B}$. Let m_+, m_-, m_0 be similarly defined with respect to $\mathcal{C}$. Then we wish to show $n_+ = m_+$, $n_- = m_-$, and $n_0 = m_0$.

Up to reordering, let $\mathcal{B}_+ = \{v_1, \ldots, v_{n_+}\}$ be the basis vectors corresponding to the positive entries along the diagonal, let $\mathcal{B}_- = \{v_{n_++1}, \ldots, v_{n_++n_-}\}$ be the basis vectors corresponding to the negative entries along the diagonal, and let the rest (denoted by $\mathcal{B}_0$) correspond to the zero entries. Similarly let $\mathcal{C}_+ = \{w_1, \ldots, w_{m_+}\}$ correspond to the positive entries, $\mathcal{C}_- = \{w_{m_++1}, \ldots, v_{m_++m_-}\}$ correspond to the negative entries, and the rest (denoted by $\mathcal{C}_0$) correspond to the zero entries.

Now use this to define the subspaces $V_+ = \operatorname{span}(\mathcal{B}_+)$, $V_{\geq 0} = \operatorname{span}(\mathcal{B}_+ \cup \mathcal{B}_0)$, and similarly define V_- and $V_{\leq 0}$. Also similarly define these subspaces for the basis $\mathcal{C}$, denoted by $W_+, W_{\geq 0}$, etc.

Now note by linearity $B(v, v) > 0$ for all $v \in V_+ \setminus \{0\}$ and $B(v, v) \geq 0$ for all $v \in V_{\geq 0}$. The analogous properties hold for the other subspaces.

Consider the subspaces V_+ and $W_{\leq 0}$. We have

$$\dim(V_+ \cap W_{\leq 0}) = \dim(V_+) + \dim(W_{\leq 0}) - \dim(V_+ + W_{\leq 0}) \geq n_+ + (n - m_+) - n = n_+ - m_+$$

Now let $v \in V_+ \cap W_{\leq 0}$. If $v \neq 0$ then we would have $B(v, v) > 0$ and $B(v, v) \leq 0$, a contradiction. Hence $v = 0$, and $\dim(V_+ \cap W_{\leq 0}) = 0$. Thus we have $n_+ \leq m_+$. Repeating this argument with W_+ and $V_{\leq 0}$, we get $n_+ \geq m_+$, and hence $n_+ = m_+$.

Now do the argument with V_- and $W_{\geq 0}$, along with W_- and $V_{\geq 0}$ to conclude $n_- = m_-$. This then forces $n_0 = m_0$. $\square$

Remark 21.2. Note that since we have a bijection between symmetric bilinear forms and quadratic forms, we could have stated this theorem for quadratic forms as well.

In the proof we explicitly constructed subspaces with dimension corresponding to the number of positive, negative, or zero entries, but furthermore, the bilinear form is respectively positive, negative, or zero on those subspaces (except possibly at 0). In essence the number of positive entries on the diagonal gives us exactly the dimension of the largest subspace on which the bilinear form is positive. This idea will shortly motivate Sylvester's Criterion for Positivity.

I also highly recommend reading the proof of this theorem from Linear Algebra Done Wrong by Treil, as it makes some of these ideas I've mentioned more explicit.

This allows for the following definition to be well-defined:

Definition 21.3 (Signature of a Symmetric Bilinear Form) — Let V be finite dimensional, $F = \mathbb{R}$, and B be a symmetric bilinear form. Given a basis $\mathcal{B} = \{v_1, \ldots, v_n\}$ for which the matrix fo B is diagonal Let n_+, n_-, n_0 respectively be the number of positive, negative, and zero entries along the diagonal. Then the triple (n_+, n_-, n_0) is called the **signature** of B.

One important notion related operators on inner product spaces and bilinear forms is positive-definiteness. It is again one of those ideas that appears in many different contexts, for example in the second derivative test from multivariable calculus. In single variable calculus you may recall a function is convex iff $f'' \geq 0$. Now it turns out in higher dimensions, the correct analogue to the second derivative is a bilinear form, but that is no longer a number, so you cannot say whether it is positive or not. Instead, the correct notion is what we will define as positive-definiteness, which gives a sufficient and necessary condition for being convex in higher dimensions.

Definition 21.4 (Definiteness) — Let V be an inner product space and $T : V \to V$ be a self-adjoint linear operator. We say T is

- **positive definite** if $\langle T(v), v\rangle > 0$ for all $v \in V \setminus \{0\}$.
- **positive semi-definite** if $\langle T(v), v\rangle \geq 0$ for all $v \in V$.
- **negative definite** if $\langle T(v), v\rangle < 0$ for all $v \in V \setminus \{0\}$.
- **negative semi-definite** if $\langle T(v), v\rangle \leq 0$ for all $v \in V$.
- **indefinite** if there exists $v, w \in V$ such that $\langle T(v), v\rangle > 0$ but $\langle T(w), w\rangle < 0$.

Let $A \in M_{n\times n}(F)$ be Hermitian. We say A is **positive definite** if $x^*Ax > 0$ for all $x \in F^n \setminus \{0\}$. The other notions are similarly defined.

Let $B : V \times V \to F$ be a symmetric bilinear form. We say B is **positive definite** if $B(v, v) > 0$ for all $v \in V \setminus \{0\}$. The other notions are similarly defined.

Let $q : V \to F$ be a quadratic form. We say q is **positive definite** if $q(v) > 0$ for all $v \in V \setminus \{0\}$. The other notions are similarly defined.

Remark 21.5. While we defined the same concept for four different kinds of objects, we can see that they are really getting at the same idea, and are interchangeable.

The notions of positive definiteness for bilinear and quadratic forms are clearly equivalent. The notions of positive definiteness of operators and matrices are also equivalent, by taking an orthonormal basis (which always exists in finite dimensions). By now considering the matrix of a bilinear form (on a finite dimensional space), it is easy to see that a bilinear form is positive definite iff its matrix is positive definite.

Note that $\langle T(x), y\rangle$ defines a bilinear form. A more subtle fact is that when the space is finite dimensional, bilinear forms are also induced by linear operators on inner product spaces, via the Riesz Representation theorem: Given a bilinear form $B(x, y)$, the linear functional φ_x defined by $y \mapsto B(x, y)$ can be written as an inner product $\varphi_x(y) = \langle z_x, y\rangle$ for some $z_x \in V$. Then you can check $T : V \to V$ defined by $T(x) = z_x$ gives $B(x, y) = \langle T(x), y\rangle$.

This does hold to an extent in infinite dimensional spaces, particularly Hilbert spaces, when the bilinear form B is "bounded."

Proposition 21.6 (Characterization of Definiteness)
Let V be a finite dimensional inner product space and let T be self-adjoint. Then

(i) T is positive definite iff every eigenvalue is positive.

(ii) T is positive semi-definite iff every eigenvalue is nonnegative.

(iii) T is negative definite iff every eigenvalue is negative.

(iv) T is negative semi-definite iff every eigenvalue is nonpositive.

(v) T is indefinite iff there are both positive and negative eigenvalues.

A similar statement holds for matrices.

Proof. We only prove (i), as the rest of the proofs are the same.

($\Longrightarrow$) Let T be positive definite. Let λ be any eigenvalue and let $v \neq 0$ be a corresponding eigenvector. Then $\langle T(v), v\rangle = \lambda\langle v, v\rangle = \lambda\|v\|^2 > 0$. Since $\|v\|^2 \geq 0$ it follows $\lambda > 0$.

($\Longleftarrow$) Now suppose every eigenvalue is positive. By the Spectral Theorem, Part I, there exists an orthonormal eigenbasis $\mathcal{B} = \{v_1, \dots, v_n\}$. Let λ_i be the eigenvalue for v_i. Then for any $v \in V \setminus \{0\}$ we can write $v = a_1v_1 + \cdots + a_nv_n$ for $a_i \in F$, so

$$\begin{aligned}
\langle T(v), v\rangle &= \left\langle T\left(\sum_{i=1}^n a_iv_i\right), \sum_{j=1}^n a_jv_j\right\rangle \\
&= \left\langle \sum_{i=1}^n \lambda_i a_i v_i, \sum_{j=1}^n a_jv_j\right\rangle \\
&= \sum_{i=1}^n\sum_{j=1}^n \lambda_i a_i \overline{a_j}\langle v_i, v_j\rangle \\
&= \sum_{i=1}^n \lambda_i\|a_i\|^2 > 0
\end{aligned}$$

□

We now state Sylvester's Criterion for Positivity:

Theorem 21.7 (Sylvester's Criterion for Positivity)
Let $A \in M_{n\times n}(F)$ be Hermitian. Define A_k to be the $k \times k$ submatrix in the top-left corner. Then A is positive definite iff $\det(A_k) > 0$ for each k.

Proof. ($\implies$) Consider the quadratic form $q : F^n \to F$ defined by $q(x) = x^T Ax$. Let $V_k := \operatorname{span}\{v_1, \ldots, v_k\}$. Then since A is positive definite, q is positive definite as well, so the restriction $q|_{V_k}$ must be as well. Then you can check the matrix of $q|_{V_k}$ in the standard basis is exactly A_k, so A_k is positive definite. Hence all the eigenvalues of A_k are positive, and since the determinant is the product of all the eigenvalues, we must have $\det(A_k) > 0$.

($\impliedby$) We induct on the dimension. This is clearly true for $n = 1$. Now assume the theorem is true for all $(n-1) \times (n-1)$ matrices. Now we can write A in block form as

$$A = \begin{pmatrix} A_{n-1} & b \\ b^* & c \end{pmatrix}$$

where $b \in F^{n-1}$ and $c \in F$. Note that A_{n-1} is also Hermitian. Since $\det(A_k) > 0$ for all $k \leq n-1$ by the induction hypothesis A_{n-1} is positive definite. Since the eigenvalues are all positive, the determinant must be nonzero, and hence A_{n-1} is invertible. We use a nifty matrix to decompose A.

$$\text{Let } S = \begin{pmatrix} I_{n-1} & -A_{n-1}^{-1}b \\ 0 & 1 \end{pmatrix}, \quad S^*AS = \begin{pmatrix} A_{n-1} & 0 \\ 0 & a_n \end{pmatrix}$$

where $a_n = c - b^* A_{n-1}^{-1} b$. Then it is easy to see $\det(S^*AS) = 1 \cdot \det(A) \cdot 1$ and $\det(S^*AS) = \det(A_{n-1}) \cdot a_n$ so it follows $a_n > 0$. Now given $x \in F^n$ we divide it into blocks:

$$x = \begin{pmatrix} y \\ z \end{pmatrix}, \, y \in F^{n-1}, \, z \in F \implies x^*Ax = y^*A_{n-1}y + z(y^*b) + \overline{z}(b^*y) + c|z|^2$$

By writing $\overline{z}(b^*y) = (zb)^* A_{n-1}^{-1} A_{n-1} y = (zbA_{n-1}^{-1})^* A_{n-1} y$ and similarly for the other cross term, we have

$$x^*Ax = (y + zA_{n-1}^{-1}b)^* A_{n-1}(y + zA_{n-1}^{-1}b) + a_n|z|^2$$

and since A_{n-1} is positive definite, the first term is positive. Since $a_n > 0$, so is the second term and we are done. □

Remark 21.8. It is slightly easier to state this theorem for matrices due to the $k \times k$ submatrices. If you wanted to state this more abstractly, you could look at a sequence of nested subspaces.

The intuition for this theorem is that the number of positive eigenvalues corresponds with the dimension of the maximal positive subspace, so if you can find subspaces of every dimension which is positive, then all the eigenvalues should be positive.

Linear Algebra Done Wrong gives two proofs of this theorem, and I think the second one is really interesting to read, just for the ideas it introduces.

Lecture 22 May 18, Generalized Eigenspaces

Let V be a finite dimensional vector space.

Now our goal is to build up to what is called the Jordan canonical form. We already saw that when an operator is diagonalizable, we get an eigenbasis, which gives us a decomposition of the space in terms of one-dimensional invariant subspaces (so in that basis the operator has a diagonal matrix). However, not every operator can be diagonalized, so we just try to do the next best thing: decompose the space into a direct sum of invariant subspace of *some* dimension.

Just accomplishing that is not too difficult: Take any basis and look at the span of the first vector. If it is an invariant subspace we are done. If not, add in the second vector, and see if the span of these two vectors is invariant, and so on. Since the basis is finite, we must eventually exhaust all the basis vectors, and since the vector space is closed under addition and scaling, we must get an invariant subspace. Once we've found an invariant subspace, we repeat this process with the remaining basis vectors, and this gives us a direct sum decomposition.

However, this is not a very informative decomposition, as some invariant subspaces could be larger than they have to be (they might be the direct sum of smaller invariant subspaces, just not obvious in a given basis), and most critically the invariant subspaces aren't associated with a nice quantity like an eigenvalue. The issue is that if we look at an operator whose characteristic polynomial splits over F, but is still non-diagonalizable, we will find that there are not enough eigenvectors to make a basis.

We introduce generalized eigenvectors and eigenspaces for this purpose. To motivate the definition, we first start with the following facts:

Proposition 22.1

Let T be a linear operator on V. Then

$$\{0\} = \ker(T^0) \subseteq \ker(T^1) \subseteq \ker(T^2) \subseteq \dots$$

Proof. Given $k \in \mathbb{N}$, if $v \in \ker(T^k)$, then $T^k(v) = 0$, so $T^{k+1}(v) = T(T^k(v)) = 0$ and hence $v \in \ker(T^{k+1})$. □

But since our space is finite dimensional, intuitively the kernels cannot keep on growing, as we would eventually exhaust all the dimensions.

Proposition 22.2

Let T be a linear operator on V. If there is some $k \in \mathbb{N}$ such that $\ker(T^k) = \ker(T^{k+1})$, then

$$\ker(T^k) = \ker(T^m) \text{ for all } m \geq k, \text{ i.e. } \ker(T^k) = \ker(T^{k+1}) = \ker(T^{k+2}) = \dots$$

Proof. Note that by induction, it suffices to show $\ker(T^m) = \ker(T^{m+1})$ for all $m \geq k$. We already have $\ker(T^m) \subseteq \ker(T^{m+1})$ so it suffices to prove the other direction. Given $v \in \ker(T^{m+1})$ we have $T^{k+1}(T^{m-k}(v)) = T^{m+1}(v) = 0$. Thus $T^{m-k}(v) \in \ker(T^{k+1}) = \ker(T^k)$. But this means $T^k(T^{m-k}(v)) = T^m(v) = 0$, and hence $T^m(v) = 0$, proving the other inclusion. □

Remark 22.3. This fact tells us two important things: 1. If the size of the kernel does not increase once, then it will never increase again. 2. Before this initial point (which exists by the well-ordering principle), the dimensions of the kernel of the powers must strictly increase.

Proposition 22.4 (Kernels of powers of T stabilize)

There exists some $k \in \mathbb{N}$ such that $\ker(T^k) = \ker(T^{k+1}) = \dots$

Proof. It suffices to show there is some $k \in \mathbb{N}$ such that $\ker(T^k) = \ker(T^{k+1})$. Suppose not. That is, for all $k \in \mathbb{N}$, we have $\ker(T^k) \subsetneq \ker(T^{k+1})$. But this means $\dim\ker(T^k) + 1 \leq \dim\ker(T^{k+1})$. (Choose a basis for $\ker(T^k)$. The strict subset means this basis does not span $\ker(T^{k+1})$, so we must at least one vector to extend it to basis of $\ker(T^{k+1})$.)

Let $\dim(V) = n$. Then by induction we have $\dim\ker(T^{n+1}) \geq \dim\ker(T^0) + n + 1 \geq n + 1$, a contradiction since $\ker(T^{n+1})$ is a subspace of V. □

Let us give one more fact which will be useful shortly, but is also a neat decomposition.

Proposition 22.5

Let T be a linear operator on V. Let $k \in \mathbb{N}$ be the smallest integer such that $\ker(T^k) = \ker(T^{k+1}) = \dots$ Then

$$V = \ker(T^k) \oplus \operatorname{im}(T^k)$$

Proof. Let $v \in \ker(T^k) \cap \operatorname{im}(T^k)$. Then $T^k(v) = 0$ and there exists $u \in V$ such that $v = T^k(u)$. Then $T^{2k}(u) = 0$. But since $\ker(T^{2k}) = \ker(T^k)$, we have $v = T^k(u) = 0$ as well, so $\ker(T^k) \cap \operatorname{im}(T^k) = \{0\}$. Now by Rank-Nullity we have

$$\begin{aligned}
\dim(\ker(T^k) + \operatorname{im}(T^k)) &= \dim(\ker(T^k)) + \dim(\operatorname{im}(T^k)) - \dim(\ker(T^k) \cap \operatorname{im}(T^k)) \\
&= \dim(\ker(T^k)) + \dim(\operatorname{im}(T^k)) \\
&= \dim(V)
\end{aligned}$$

It follows $V = \ker(T^k) + \operatorname{im}(T^k)$, and we are done. □

We apply this idea to a specific operator, namely $T - \lambda\mathbf{1}_V$, since the eigenspaces are exactly defined as $E_\lambda(T) = \ker(T - \lambda\mathbf{1}_V)$.

Definition 22.6 (Generalized Eigenvectors and Eigenspaces) — Let T be a linear operator on V and let λ be an eigenvalue of T. A nonzero vector $v \in V$ is called a **generalized eigenvector** of T corresponding to λ if there exists some positive integer k such that $(T - \lambda\mathbf{1}_V)^k(v) = 0$.

We define the **generalized eigenspace** of T corresponding λ by

$$G_\lambda(T) := \{v \in V : (T - \lambda \mathbf{1}_V)^k(v) = 0 \text{ for some positive integer } k\}$$

i.e. $G_\lambda(T)$ is the set of all the generalized eigenvectors corresponding to λ along with 0.

Remark 22.7. Alternatively, we can write

$$G_\lambda(T) = \bigcup_{k=1}^{\infty} \ker(T - \lambda \mathbf{1}_V)^k$$

But since we showed there is some number m for which the kernel stabilizes, we can actually write this as a finite union or even one kernel by the nestedness property:

$$G_\lambda(T) = \bigcup_{k=1}^{m} \ker(T - \lambda \mathbf{1}_V)^k = \ker(T - \lambda \mathbf{1}_V)^m$$

Theorem 22.8

Let T be a linear operator on V, and suppose the characteristic polynomial of T splits over F. Then there is a basis of V consisting of generalized eigenvectors of T.

Proof. Let $\dim(V) = n$. We induct on the dimension. This is trivially true when $n = 1$ since every vector is an eigenvector. Now suppose the result holds for operators on spaces of all smaller dimensions. Choose k as in Proposition 22.5, so we have $V = \ker(T - \lambda \mathbf{1}_V)^k \oplus \operatorname{im}(T - \lambda \mathbf{1}_V)^k$.

Let λ be an eigenvalue of T (since the characteristic polynomial splits), and thus there is a nonzero eigenvector, so $\ker(T - \lambda \mathbf{1}_V)^k \neq \{0\}$. Then by Rank-Nullity this implies $\operatorname{im}(T - \lambda \mathbf{1}_V)^k < n$. Note that since polynomials of T commute with T, $\operatorname{im}(p(T))$ is T-invariant for any polynomial p. (If for any $v \in \operatorname{im}(p(T))$ $v = p(T)(u)$ we have $T(v) = T(p(T)(u)) = p(T)(T(u))$.) Taking $p(t) = (t - \lambda)^k$, we get that $\operatorname{im}(T - \lambda \mathbf{1}_V)^k$ is T-invariant.

Then the operator $T|_{\operatorname{im}(T - \lambda \mathbf{1}_V)^k}$ is well-defined, and its characteristic polynomial must also split (since it divides the characteristic polynomial of T). Then by the induction hypothesis, there is a basis for $\operatorname{im}(T - \lambda \mathbf{1}_V)^k$ consisting of generalized eigenvectors for the restricted operator. But these must also be generalized eigenvectors for T.

Now take any basis of $\ker(T - \lambda \mathbf{1}_V)^k$ (this space by definition contains only generalized eigenvectors and 0), and joining this to the basis we have for $\operatorname{im}(T - \lambda \mathbf{1}_V)^k$, we get a full basis of V by the direct sum. $\square$

Remark 22.9. The above proof is from Axler. This proof can also be done with quotient spaces in a similar spirit to the proofs we have done before: We argue by induction on dimension. Pick an eigenvalue and corresponding eigenvector, and consider the span of that vector, say $\operatorname{span}(v)$. Since it is an invariant subspace, this induces an operator $\overline{T} : V/\operatorname{span}(v) \to V/\operatorname{span}(v)$.

> Then, apply the inductive hypothesis to this operator to get a basis of generalized eigenvectors in the quotient space. Pulling this back via the quotient map, we can check this gives us a list of generalized eigenvectors in the original space V. Then adding the eigenvector v gives us a whole basis of generalized eigenvectors for V.

Proposition 22.10

Each $G_\lambda(T)$ is T-invariant.

Proof. Let λ be an eigenvalue of T. Let $v \in G_\lambda(T)$, so $(T - \lambda \mathbf{1}_V)^k(v) = 0$ for some positive integer k. Now because polynomials of T commute with T, we have that $\ker(p(T))$ is T-invariant for any polynomial p (if $p(T)(v) = 0$, then $p(T)(T(v)) = T(p(T)(v)) = 0$). Then taking $p(t) = (t - \lambda)^k$, we have that $(T - \lambda \mathbf{1}_V)^k(T(v)) = 0$ as well, so $T(v) \in G_\lambda(T)$, and we are done. □

The above two facts combined tell us that the concept of generalized eigenspaces we define do indeed give us the invariant subspaces we desire, and they also give us enough vectors to form a full basis. Next time we will show that they indeed give us a direct sum decomposition like we had with diagonalizable operators, and after that we will build up more properties to show why this particular decomposition is useful.

Lecture 23 May 20, Generalized Eigenspace Decomposition

Let V be a finite dimensional vector space.

We show that these generalized eigenspaces give us a direct sum decomposition of V.

Theorem 23.1

Let T be a linear operator on V. Let $\lambda, \mu \in F$ be eigenvalues. Then if $\lambda \neq \mu$ we have $G_\lambda(T) \cap G_\mu(T) = \{0\}$.

Proof. Suppose $v \in V$ is a (nonzero) generalized eigenvector of T corresponding to λ and μ. Then let k be the smallest positive integer such that $(T - \mu\mathbf{1}_V)^k(v) = 0$. Then

$$\begin{aligned} 0 &= (T - \lambda\mathbf{1}_V)^k(v) \\ &= \left((T - \mu\mathbf{1}_V) + (\mu - \lambda)\mathbf{1}_V\right)^n(v) \\ &= \sum_{i=0}^{n} \binom{n}{k} (\mu - \lambda)^{n-k} (T - \mu\mathbf{1}_V)^k(v) \end{aligned}$$

By multiplying everything by $(T - \mu\mathbf{1}_V)^{k-1}$, the only term left is the first term $0 = (\mu - \lambda)^n(T - \mu\mathbf{1}_V)^{k-1}(v)$. But since $(T - \mu\mathbf{1}_V)^{k-1}(v) \neq 0$, it follows $\mu = \lambda$, a contradiction. Thus the only element in the intersection is 0. $\square$

Proposition 23.2

Let T be a linear operator on V. If $v_1, \ldots, v_m$ are generalized eigenvectors corresponding to distinct eigenvalues $\lambda_1, \ldots, \lambda_m$, then they are linearly independent.

Proof. Suppose the result is false. Then there exists (by the well-ordering principle) a smallest m for which there is a list of linearly dependent generalized eigenvectors $v_1, \ldots, v_m$ corresponding to distinct eigenvalues $\lambda_1, \ldots \lambda_m$. Then there exists $a_1, \ldots, a_m \in F$ (which we can assume to be nonzero by the minimality of m) such that

$$a_1 v_1 + \cdots + a_m v_m = 0$$

To simplify notation, note that we can choose a number n sufficiently large such that $(T - \lambda_i\mathbf{1}_V)^n(v_i) = 0$ for each i (do this by taking the max of all the exponents needed for each generalized eigenvector v_i). Then, by applying $(T - \lambda_m\mathbf{1}_V)^n$ to each side, we have

$$a_1(T - \lambda_m\mathbf{1}_V)^n(v_1) + \cdots + a_{m-1}(T - \lambda_{m-1}\mathbf{1}_V)^n(v_{m-1}) = 0$$

Note none of these terms can be zero, since each generalized eigenvector cannot correspond to two distinct eigenvalues. But now note that since polynomials of T commute, for each i we have

$$(T - \lambda_i\mathbf{1}_V)^n((T - \lambda_m\mathbf{1}_V)^n(v_i)) = (T - \lambda_m\mathbf{1}_V)^n((T - \lambda_i\mathbf{1}_V)^n(v_i)) = 0$$

Hence $(T - \lambda_m\mathbf{1}_V)^n(v_i)$ is a generalized eigenvector corresponding to λ_i for each i. But this means we a linearly dependent list of generalized eigenvectors corresponding to distinct eigenvalues of length $m - 1$, a contradiction to the minimality of m. $\square$

Theorem 23.3 (Generalized Eigenspace Decomposition)
Let T be a linear operator on V, and suppose the characteristic polynomial of T splits over F. Let $\lambda_1, \dots, \lambda_m$ be the distinct eigenvalues of T. Then

$$V = G_{\lambda_1}(T) \oplus \cdots \oplus G_{\lambda_m}(T)$$

Proof. We have a basis of generalized eigenvectors, so by grouping them according to their generalized eigenspace, for any $v \in V$ we can write $v = w_1 + \cdots + w_m$, where $w_i \in G_{\lambda_i}(T)$. To show uniqueness, it suffices to show that if $u_1 + \cdots + u_m = 0$, for some $u_i \in G_{\lambda_i}(T)$, then each $u_i = 0$. (See 115AH HW4.) But this must be true, since generalized eigenvectors corresponding to distinct eigenvalues must be linearly independent. Hence we have a direct sum. □

Lecture 24 May 22, Nilpotent Operators, Cycles, and Jordan Canonical Form

Let V be a finite dimensional vector space over F.

Now that we have a nice direct sum decomposition in terms of generalized eigenspaces, we would like to investigate the generalized eigenspaces themselves further. In particular we would like to know if there is some nice basis for which the matrix of the operator has a relatively simple form, which is what will lead us to Jordan Canonical Form. To do this we will use nilpotent operators.

Nilpotent Operators

Definition 24.1 (Nilpotent operator) — Let T be a linear operator on V (not necessarily finite dimensional). We say T is nilpotent if there exists some $k \in \mathbb{N}$ such that $T^k = 0$.

Proposition 24.2

Let T be a linear operator on V.

(i) If T is nilpotent, then 0 is an eigenvalue and also the only eigenvalue.

(ii) If the characteristic polynomial of T splits over F, and 0 is the only eigenvalue of T, then T is nilpotent.

Proof. (i) Let $k \in \mathbb{N}$ be the smallest number such that $T^k = 0$. Then $T^{k-1} \neq 0$ so there exists some nonzero $v \in \operatorname{im}(T^{k-1})$. Then $T(v) = 0$, so 0 is an eigenvalue. If λ is an eigenvalue, let v be an eigenvector, so $T(v) = \lambda v$. Inducing, we have $T^k(v) = \lambda^k v = 0$, so $\lambda = 0$.

(ii) If the only eigenvalue is 0, the characteristic polynomial cannot have any roots other than 0, and since it splits over F, we must have $f_T(t) = t^n$ where $n = \dim(V)$. Then by Cayley-Hamilton we have $T^n = 0$.

□

This gives us an immediate corollary:

Corollary 24.3

Let T be a linear operator on V, and suppose the characteristic polynomial of T splits over F. If T is nilpotent, then its characteristic polynomial is given by $f_T(t) = t^n$ where $n = \dim(V)$.

In particular, one important operator is nilpotent:

Proposition 24.4

Let T be a linear operator on V, λ be an eigenvalue, and let $G_\lambda(T)$ be the corresponding generalized eigenspace. Then $(T - \lambda \mathbf{1}_V)|_{G_\lambda(T)}$ is nilpotent.

Proof. Let $m \in \mathbb{N}$ be the smallest integer such that the kernels stabilize, i.e. $\ker(T - \lambda \mathbf{1}_V)^m = \ker(T - \lambda \mathbf{1}_V)^{m+1} = \dots$ Then for each $v \in G_\lambda(T)$ we must have $(T - \lambda \mathbf{1}_V)^m(v) = 0$. Thus $(T - \lambda \mathbf{1}_V)^m|_{G_\lambda(T)} = 0$. □

We now give an important characterization of the dimension of generalized eigenspaces:

Theorem 24.5

Let T be a linear operator on V whose characteristic polynomial splits, and let $\lambda \in F$ be an eigenvalue of T. Let k be the algebraic multiplicity of λ, so the characteristic polynomial of T is given by $f_T(t) = (t - \lambda)^k g(t)$ for some polynomial g. Then $G_\lambda(T) = \ker(T - \lambda \mathbf{1}_V)^k$ and $\dim G_\lambda(T) = k$.

Proof. Let $m \in \mathbb{N}$ be the smallest integer such that the kernels stabilize, i.e. $\ker(T - \lambda \mathbf{1}_V)^m = \ker(T - \lambda \mathbf{1}_V)^{m+1} = \dots$ Since $(T - \lambda \mathbf{1}_V)|_{G_\lambda(T)}$ is nilpotent, its characteristic polynomial is t^d where $d = \dim G_\lambda(T)$. Now we have that the characteristic polynomial of $T|_{G_\lambda(T)}$ is given by

$$f_{T|_{G_\lambda(T)}}(t) = \det(T|_{G_\lambda(T)} - t\mathbf{1}_{G_\lambda(T)}) = \det((T - \lambda \mathbf{1}_V)|_{G_\lambda(T)} - (t - \lambda)\mathbf{1}_{G_\lambda(T)}) = (t - \lambda)^d$$

But since the characteristic polynomial of $T|_{G_\lambda(T)}$ divides the characteristic polynomial of T, it follows $d \leq k$. But this must be true for each eigenvalue, and since the characteristic polynomial of T splits, we must have that all the algebraic multiplicities sum to $n = \dim(V)$. But by the generalized eigenspace decomposition, the dimensions of the generalized eigenspaces must also sum to n. But with the bound $d \leq k$, the only way this can happen is if $d = k$, so $\dim G_\lambda(T) = k$.

Since m is the smallest integer for which the kernels stabilize, as noted before, the dimension of the kernels before this must strictly increase, i.e.

$$0 = \dim \ker(T - \lambda \mathbf{1}_V)^0 < \dim \ker(T - \lambda \mathbf{1}_V)^1 < \dots < \dim \ker(T - \lambda \mathbf{1}_V)^m$$

Then it follows $\dim G_\lambda(T) = \dim \ker(T - \lambda \mathbf{1}_V)^m \geq m$. Putting this together with what we have above, we get $m \leq k$, so

$$G_\lambda(T) = \ker(T - \lambda \mathbf{1}_V)^m = \ker(T - \lambda \mathbf{1}_V)^k$$

□

Recall that an operator is diagonalizable iff the algebraic multiplicity of each eigenvalue is equal to its geometric multiplicity. This immediately gives us the following corollary:

Corollary 24.6

Let T be a linear operator on V, and suppose the characteristic polynomial of T splits over F. Then T is diagonalizable iff for each eigenvalue λ we have $\dim G_\lambda(T) = \dim E_\lambda(T)$.

Cycles of Generalized Eigenvectors

Now we attempt to find a "nice" basis for each generalized eigenspace.

Definition 24.7 (Cycle of generalized eigenvectors) — Let T be a nilpotent operator on V. We call an ordered set $\{v_k, \dots, v_1\} \subseteq V$ a **cycle of generalized eigenvectors** if

$$T(v_1) = 0 \quad \text{and} \quad T(v_{j+1}) = v_j$$

v_1 is called the **initial vector** and v_k is called the **end vector**.

Remark 24.8. Note that the 0 is the only eigenvalue, so v_1 is the only true eigenvector. Then the rest of the vectors are generalized eigenvectors since $0 = T^j(v_j) = (T - 0\mathbf{1}_V)^j(v_j)$.

This definition seems a little arbitrary at the moment, but this is because we haven't explicitly said what the "nice" basis should look like. At the end we will go back and revisit everything we have done in terms of matrices, which will actually give some motivation for how we defined certain things.

Example 24.9

If T is any linear operator on V, then $(T - \lambda\mathbf{1}_V)|_{G_\lambda(T)}$ is nilpotent. Then any ordered set of vectors $\{v_k, \dots, v_1\} \subseteq G_\lambda(T)$ is a cycle of generalized eigenvectors if

$$(T - \lambda\mathbf{1}_V)(v_1) = 0 \quad \text{and} \quad (T - \lambda\mathbf{1}_V)(v_{j+1}) = v_j$$

Theorem 24.10

Let T be a nilpotent operator on V, and let $\mathcal{C}_1, \dots, \mathcal{C}_m$ be cycles of generalized eigenvectors, with $\mathcal{C}_i = \{v^i_{k_i}, \dots, v^i_1\}$, so each cycle has length $k_i \in \mathbb{N}$. Let $\mathcal{C} = \mathcal{C}_1 \cup \dots \cup \mathcal{C}_m$. Suppose the initial vectors $v^1_1, \dots, v^m_1$ are linearly independent. Then all the cycles are pairwise disjoint, and their union $\mathcal{C}$ is linearly independent.

Proof. Let $n = k_1 + \dots + k_m$ be the total number of vectors in $\mathcal{C}$. We induct on n. The base case is trivial.

Now assume the theorem is true for any operator and any collection of cycles as long as their total length is less than n. It is easy to check $\operatorname{span}(\mathcal{C}_i)$ is T-invariant, and thus so is $\operatorname{span}(\mathcal{C})$. Then WLOG assume $\mathcal{C}$ spans all of V, because otherwise we can always restrict T to $\operatorname{span}(\mathcal{C})$.

Since $\mathcal{C}$ spans V, by the relations that define the cycles, we must have that $\bigcup_{i=1}^m \mathcal{C}_i \setminus \{v^i_{k_i}\}$ spans $\operatorname{im}(T)$ (where we just remove the end vector from each cycle). Then by the inductive hypothesis, this set of vectors is linearly independent, and are thus a basis. Thus we have $\operatorname{rank}(T) = n - m$.

On the other hand we have $T(v^i_1) = 0$ for each i, and since we assumed the initial vectors to be linearly independent, we have $\dim\ker(T) \geq m$. Thus by Rank-Nullity we have $\dim(V) = \dim\ker(T) + \operatorname{rank}(T) \geq n$. But since $\mathcal{C}$ is a spanning set of length n, it follows it must be a basis, and thus linearly independent. □

Theorem 24.11
Let T be a nilpotent operator on V. Then V has a basis consisting of unions of cycles of generalized eigenvectors.

Proof. We induct on the dimension $n = \dim(V)$. The base case is trivial. Assume the result is true for any operator on any space with less than n dimensions.

Consider the invariant subspace $\mathrm{im}(T)$ and the restriction $T|_{\mathrm{im}(T)}$. Since T is nilpotent, 0 is an eigenvalue, and this means T is not invertible. Hence $\mathrm{rank}(T) < n$, so by the induction hypothesis, there exists cycles $\mathcal{C}_1, \ldots, \mathcal{C}_m$, each given by $\mathcal{C}_i = \{v^i_{k_i}, \ldots, v^i_1\}$, such that their union is a basis for $\mathrm{im}(T)$.

By definition of the cycles we have $v^i_1 \in \ker(T)$, and the initial vectors are linearly independent since they are part of a basis. Thus extend this to a basis $\{v^1_1, \ldots, v^m_1, u_1, \ldots, u_r\}$ for $\ker(T)$. We can consider each u_j as a length 1 cycle of generalized eigenvectors. Thus $\dim\ker(T) = m + r$.

But we used some of the vectors in the basis of $\mathrm{im}(T)$ as part of the basis for $\ker(T)$, so we need to compensate. Since each end vector $v^i_{k_i} \in \mathrm{im}(T)$ there exists some $v^i_{k_i+1}$ such that $T(v^i_{k_i+1}) = v^i_{k_i}$, so we can extend each cycle by one. Let $\tilde{\mathcal{C}}_i = \{v^i_{k_i+1}, v^i_{k_i}, \ldots, v^i_1\}$. Thus their dimensions must sum up to $\mathrm{rank}(T) + m$.

Now we claim $\tilde{\mathcal{C}}_1 \cup \cdots \cup \tilde{\mathcal{C}}_m \cup \{u_1, \ldots, u_r\}$ is a basis for V. This is a union of cycles whose initial vectors $\{v^1_1, \ldots, v^m_1, u_1, \ldots, u_r\}$ are linearly independent, so by the previous theorem, the union is linearly independent. To see it is a basis, we count dimensions. We have

$$\dim\mathrm{span}(\tilde{\mathcal{C}}_1 \cup \cdots \cup \tilde{\mathcal{C}}_m \cup \{u_1, \ldots, u_r\}) = (\mathrm{rank}(T) + m) + r = \mathrm{rank}(T) + \dim\ker(T) = \dim(V)$$

□

Jordan Canonical Form

Definition 24.12 (Jordan canonical basis) — Let T be a linear operator on V. We say a basis $\mathcal{B}$ for V is a **Jordan canonical basis** such that the matrix of T with respect to $\mathcal{B}$ has block diagonal form

$$\begin{pmatrix} A_1 & & \\ & \ddots & \\ & & A_m \end{pmatrix}$$

where each block matrix is given by

$$A_i = \begin{pmatrix} \lambda_i & 1 & & \\ & \lambda_i & \ddots & \\ & & \ddots & 1 \\ & & & \lambda_i \end{pmatrix}$$

for some $\lambda_i \in F$. (where the 1×1 block is just 0.)

Corollary 24.13
Let T be a nilpotent operator on V. Then there is a Jordan canonical basis for V.

Proof. As in the proof of Theorem 24.11, each of the cycles $\tilde{\mathcal{C}}_i$ satisfy $T(v_1^i) = 0$ and $T(v_{j+1}^i) = v_j^i$ for $1 \leq j \leq k_i$. Furthermore $T(u_j) = 0$. Writing out the matrix of T in the ordered basis $\tilde{\mathcal{C}}_1 \cup \cdots \cup \tilde{\mathcal{C}}_m \cup \{u_1, \ldots, u_r\}$, we get

$$[T]_{\mathcal{B}} = \begin{pmatrix} \left.\boxed{\begin{matrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{matrix}}\right\} k_1 & & & & & \\ & \ddots & & & & \\ & & \left.\boxed{\begin{matrix} 0 & 1 \\ 0 & 0 \end{matrix}}\right\} k_m & & & \\ & & & 0 & & \\ & & & & \ddots & \\ & & & & & 0 \end{pmatrix} \left.\vphantom{\begin{matrix} 0 \\ \ddots \\ 0 \end{matrix}}\right\} r$$

□

Theorem 24.14 (Jordan canonical form)
Let T be a linear operator on V. If the characteristic polynomial of T splits over F, then there is a Jordan canonical basis for T.

Proof. Let $\lambda_1, \ldots, \lambda_m$ be all the distinct eigenvalues of T. We have the the generalized eigenspace decomposition

$$V = G_{\lambda_1}(T) \oplus \cdots \oplus G_{\lambda_m}(T)$$

Then each $(T - \lambda_i \mathbf{1}_V)|_{G_{\lambda_i}(T)}$ is nilpotent and thus has a Jordan canonical basis. Then this is also a Jordan canonical basis for $T|_{G_{\lambda_i}(T)}$ since the matrix of $\lambda_i \mathbf{1}_V$ in any basis is just the diagonal matrix with λ_i along the diagonal. Then join the Jordan canonical basis for each generalized eigenspace to get a basis for all of V, for which the matrix is the block diagonal matrix where each block is the matrix for each $T|_{G_{\lambda_i}(T)}$. Thus this full basis is a Jordan canonical basis. □

Remark 24.15. Now if you look at the matrices of $T - \lambda \mathbf{1}_V$ restricted to each block, you see we get these nilpotent matrices, which exactly satisfy the relations of the cycles of generalized eigenvectors ($T(v_1) = 0$ and $T(v_{j+1}) = v_j$).

An alternative way to approach the Jordan canonical form would be to first ask how we can find a basis for which the matrix is block diagonal (decompose into invariant subspaces), and then use Schur's to put each block into upper triangular form. But by using generalized eigenspaces, we guarantee the diagonal will be the eigenvalue λ. Then we can notice that after we subtract

> the diagonal, we get a nilpotent matrix (you can check that by taking powers, the upper triangle gets "squeezed out"). Finally we ask if there is a nice basis such that the nilpotent matrices have the above form, where there is a diagonal of 1's above the main diagonal, which we accomplish by using the cycles of generalized eigenvectors.

Lecture 25 May 25

Holiday.

Lecture 26 May 27, Minimal Polynomial

While the characteristic polynomial is closely linked to diagonalizability, the better tool to deal with Jordan form is the minimal polynomial. We first need a result to justify the existence and uniqueness of the minimal polynomial.

Let V be a vector space over F.

Proposition 26.1

Let $T : V \to V$ be a linear operator and suppose there is at least one polynomial q such that $q(T) = 0$. Then there exists a unique monic polynomial of the lowest degree p such that $p(T) = 0$, and for any polynomial r such that $r(T) = 0$, we have $p(t) \mid r(t)$.

Proof. If we look at the set of all possible degrees of polynomials q such that $q(T) = 0$, by the well-ordering principle, there is one of minimal degree. Let $p(t)$ be such a polynomial. Suppose there is another such polynomial $\tilde{p}(t)$. WLOG assume they are both monic (by dividing by the leading coefficient). Then $(p - \tilde{p})(T) = 0$ but $p - \tilde{p}$ has strictly smaller degree than p so we must have $p - \tilde{p} = 0$. Hence p is unique.

Now let r be another polynomial such that $r(T) = 0$. By the minimality of p we must have $\deg p \leq \deg r$. By division with remainder we can write $r(t) = p(t) \cdot m(t) + s(t)$ where $\deg s < \deg p$. But then $s(T) = r(T) - m(T)p(T) = 0$, so we must have $s(t) = 0$ by the minimality of p. Hence $p(t) \mid r(t)$. □

Remark 26.2. In particular, when V is finite dimensional, the characteristic polynomial is well-defined, and so we have the Cayley-Hamilton theorem, which says $f_T(T) = 0$. Thus the minimal polynomial always exists when V is finite dimensional.

Definition 26.3 (Minimal polynomial) — The **minimal polynomial** is (if it exists) defined as the unique monic polynomial of lowest degree p_T such that $p_T(T) = 0$.

Remark 26.4. The minimal polynomial usually does not exist in infinite dimensions. For example, consider the derivative operator on the space of polynomials - you can always find larger and larger degree polynomials which are not cancelled out by the highest order derivative in your polynomial. However, like the characteristic polynomial, the minimal polynomial is not a particularly useful notion for infinite dimensional spaces, so we make the following assumption:

For the remainder of this section, let V be a finite dimensional space.

Proposition 26.5

Let $T : V \to V$ be a linear operator, and let W be an invariant subspace, $T|_W$ is well-defined. Then the $p_{T|_W} \mid p_T$.

We omit the proof.

The minimal polynomial does not have a straightforward method of computation unlike the characteristic polynomial, so we need to be more clever. The following fact is immediate and very useful:

Proposition 26.6

The minimal polynomial always divides the characteristic polynomial.

So a rather primitive method to find the minimal polynomial is to test all the polynomials dividing the characteristic polynomial until we find one equal to 0. We will later use Jordan form to provide a better method for computing the minimal polynomial.

Example 26.7 (Minimal polynomial of Jordan blocks)

Consider the following matrices:

$$A_1 = \begin{pmatrix} 2 & & \\ & 2 & \\ & & 2 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 2 & 1 & \\ & 2 & \\ & & 2 \end{pmatrix}, \quad A_3 = \begin{pmatrix} 2 & 1 & \\ & 2 & 1 \\ & & 2 \end{pmatrix}$$

The characteristic polynomial for all three matrices is $(t-2)^3$, but we can check the corresponding minimal polynomials are $p_1(t) = t-2$, $p_2(t) = (t-2)^2$, $p_3(t) = (t-2)^3$. We will later show this is not a coincidence, and in fact the minimal polynomial gives the sizes of the largest Jordan blocks of each eigenvalue.

Theorem 26.8

Let $T : V \to V$ be a linear operator and $p(t)$ be its minimal polynomial. Then $\lambda \in F$ is an eigenvalue of T iff it is a root of the minimal polynomial. Hence the minimal polynomial and the characteristic polynomial have the same roots.

Proof. ($\Longrightarrow$) If $\lambda \in F$ is an eigenvalue, let v be an eigenvector so $T(v) = \lambda v$. But then we have $0 = p(T)(v) = p(\lambda)v$, so $p(\lambda) = 0$.

($\Longleftarrow$) Now suppose $p(\lambda) = 0$. Since the minimal polynomial divides the characteristic polynomial f_T, we must have $f_T(\lambda) = 0$ so λ is an eigenvalue. $\square$

Theorem 26.9

Let $T : V \to V$ be a linear operator. T is diagonalizable iff $p_T(t) = (t - \lambda_1) \cdots (t - \lambda_k)$, where $\lambda_1, \ldots, \lambda_k$ are distinct eigenvalues, i.e. p_T splits into linear factors with no repeated roots.

Proof. ($\implies$) Suppose T is diagonalizable, so there exists an eigenbasis for V. Let $\lambda_1, \ldots, \lambda_k$ be all the distinct eigenvalues of T. Then $(T - \lambda_1 \mathbf{1}_V) \cdots (T - \lambda_k \mathbf{1}_V)(v) = 0$ for all $v \in V$ since we can write each v as a linear combination of eigenvectors, and each eigenvectors is cancelled out by one of the factors above. Thus the the minimal polynomial divides the above product, but on the other hand, since all the eigenvalues must be roots of the minimal polynomial, we must have that the minimal polynomial is exactly $p_T(t) = (t - \lambda_1) \cdots (t - \lambda_k)$.

($\impliedby$) Now suppose we have $p_T(t) = (t - \lambda_1) \cdots (t - \lambda_k)$. We induct on k, the number of distinct eigenvalues. The base case is true since $T - \lambda_1 \mathbf{1}_V = 0 \implies T(v) = \lambda_1 v$.

Now suppose the result is true for any number of distinct eigenvalues $< k$. The strategy is to split V into two smaller invariant subspaces with minimal polynomial that has strictly smaller degree, namely $\operatorname{im}(T - \lambda_k \mathbf{1}_V)$ and $\ker(T - \lambda_k \mathbf{1}_V)$.

Given $v \in \operatorname{im}(T - \lambda_k \mathbf{1}_V)$, there exists $u \in V$ such that $(T - \lambda_k \mathbf{1}_V)(u) = v$. Then by definition of the minimal polynomial we have $(T - \lambda_1 \mathbf{1}_V) \cdots (T - \lambda_{k-1} \mathbf{1}_V)(T - \lambda_k \mathbf{1}_V)(u) = 0$. Hence the minimal polynomial of $T|_{\operatorname{im}(T - \lambda_k \mathbf{1}_V)}$ must divide $(t - \lambda_1) \cdots (t - \lambda_{k-1})$, and by the induction hypothesis there is a basis of $\operatorname{im}(T - \lambda_k \mathbf{1}_V)$ which are eigenvectors of the restricted operator, and hence eigenvectors of T.

By definition $\ker(T - \lambda_k \mathbf{1}_V)$ is an eigenspace, so any basis is an eigenbasis.

We now claim $V = \operatorname{im}(T - \lambda_k \mathbf{1}_V) \oplus \ker(T - \lambda_k \mathbf{1}_V)$. If $v \in \operatorname{im}(T - \lambda_k \mathbf{1}_V) \cap \ker(T - \lambda_k \mathbf{1}_V)$, then $T(v) = \lambda_k v$. Being in $\operatorname{im}(T - \lambda_k \mathbf{1}_V)$ then implies

$$0 = (T - \lambda_1 \mathbf{1}_V) \cdots (T - \lambda_{k-1} \mathbf{1}_V)(v) = (\lambda_k - \lambda_1) \cdots (\lambda_k - \lambda_{k-1}) v = 0$$

Since all the eigenvalues are distinct, it follows $v = 0$, so the intersection is trivial. By Rank-Nullity we must also have $\dim \operatorname{im}(T - \lambda_k \mathbf{1}_V) + \dim \ker(T - \lambda_k \mathbf{1}_V) = \dim V$, so we have the desired direct sum.

Finally, we take the union of the basis of the two subspaces, which gives us an eigenbasis of V, and hence T is diagonalizable. □

Lecture 27 May 29, Minimal Polynomial and Jordan Form

Let V be a finite dimensional vector space.

Proposition 27.1

Let $T : V \to V$ be a linear operator. Suppose V is T-cyclic, so there exists $v \in V$ such that $V = \operatorname{span}\{v, T(v), T^2(v), \dots\}$. Then $p_T(t) = (-1)^n f_T(t)$.

Remark 27.2. I leave this result in because Professor Gao included it, but in context of the textbook, this seems more relevant for rational canonical form.

Proof. Let $\dim(V) = n$. Then recall from Theorem 7.5 that $\{v, T(v), \dots, T^{n-1}(v)\}$ is a basis for V. Let $g(t) = a_0 + a_1 t + \cdots + a_k t^k$ be any polynomial with $\deg g \leq n-1$. Then $g(T)(v) = a_0 v + a_1 T(v) + \cdots + a_k T^k(v) \neq 0$ since $v, \dots, T^k(v)$ are linearly independent, and $a_k \neq 0$. Hence $g(T) \neq 0$, so the minimal polynomial of T must be at least degree n. But since it divides the characteristic polynomial, it must be exactly degree n. Hence we can write $p_T(t) = c f_T(t)$ for some constant c. Since we defined the minimal polynomial to be monic, i.e. have leading coefficient 1, comparing this to the leading coefficient of the characteristic polynomial gives $c = (-1)^n$. $\square$

Now we return to Jordan canonical form to develop some of the results Gao did not cover in lecture. This will give us a way to compute Jordan form even for large matrices, and will also show that Jordan form is unique up to reordering of the blocks, which makes the last result connecting back to the minimal polynomial well-posed.

Definition 27.3 (Dot diagram) — Let $T : V \to V$ be a linear operator and let the characteristic polynomial of T split over F. Let λ_i be some eigenvalue, and consider the generalized eigenspace $G_{\lambda_i}(T)$.

The generalized eigenspace has a basis which is a union of cycles of generalized eigenvectors corresponding to λ_i. Write the vectors of cycles in columns, and arrange the columns from longest to shortest, with longest on the left. Then put a dot for each generalized eigenvector, and this is the **dot diagram** for $T|_{G_{\lambda_i}(T)}$:

$$
\begin{array}{llll}
\bullet\, (T-\lambda_i \mathbf{1}_V)^{p_1-1}(v_1) & \bullet\, (T-\lambda_i \mathbf{1}_V)^{p_2-1}(v_2) & \cdots & \bullet\, (T-\lambda_i \mathbf{1}_V)^{p_{n_i}-1}(v_{n_i}) \\
\bullet\, (T-\lambda_i \mathbf{1}_V)^{p_1-2}(v_1) & \bullet\, (T-\lambda_i \mathbf{1}_V)^{p_2-2}(v_2) & & \bullet\, (T-\lambda_i \mathbf{1}_V)^{p_{n_i}-2}(v_{n_i}) \\
\vdots & \vdots & & \vdots \\
 & \bullet\, (T-\lambda_i \mathbf{1}_V)(v_2) & & \bullet\, (T-\lambda_i \mathbf{1}_V)(v_{n_i}) \\
\bullet\, (T-\lambda_i \mathbf{1}_V)(v_1) & \bullet\, v_2 & & \bullet\, v_{n_i} \\
\bullet\, v_1 & & &
\end{array}
$$

Here there are n_i cycles, each of length p_j, and the end vectors are denoted $v_1, \dots, v_{n_i}$.

Recall that in the proof of the existence of Jordan form, we essentially showed each cycle of generalized eigenvectors corresponds exactly to a Jordan block, with the length corresponding to the size of the Jordan block. Thus the dot diagram is a nice bookkeeping device for finding all the Jordan blocks. We will show that the dot diagram is in fact unique, which implies the statement about Jordan form being unique up to reordering.

We are now interested in computing the dot diagram. We make the following observation that will guide our intuition:

> **Remark 27.4.** Note the vector at the top of each column (i.e. in the first row) is an initial vector, and hence the only true eigenvector. This means applying $T - \lambda_i \mathbf{1}_V$ once will annihilate these vectors. Similarly, for the vectors in the second row, applying $(T - \lambda_i \mathbf{1}_V)^2$ will annihilate them, and so on.

Theorem 27.5

The vectors in the first r rows of the dot diagram are a basis for $\ker(T - \lambda_i \mathbf{1}_V)^r$, and hence the number of dots in the first r rows is $\dim \ker(T - \lambda_i \mathbf{1}_V)^r$.

Proof. Note that $\ker(T - \lambda_i \mathbf{1}_V)^r \subseteq G_{\lambda_i}(T)$. Hence if we let $S = (T - \lambda_i \mathbf{1}_V)^r|_{G_{\lambda_i}(T)}$ we have $\ker(T - \lambda_i \mathbf{1}_V)^r = \ker(S)$. Thus it suffices to show the vectors are a basis for $\ker(S)$.

Let $\mathcal{B}_i$ be a basis of $G_{\lambda_i}(T)$ which is a union of cycles of generalized eigenvectors. Then define $\mathcal{C}_1 = \mathcal{B}_i \cap \ker(S)$ and $\mathcal{C}_2 = \mathcal{B}_i \setminus \ker(S)$.

Note that applying S to any vector in $\mathcal{C}_2$ moves it up r rows in the dot diagram. Hence S is injective on $\mathcal{C}_2$, so $S(\mathcal{C}_2)$ is also linearly independent. But since S maps $\text{span}(\mathcal{C}_1)$ to $\{0\}$, $S(\text{span}(\mathcal{C}_2))$ must span $\text{im}(S)$, and hence $S(\mathcal{C}_2)$ is a basis for $\text{im}(S)$. Hence $\text{rank}(S) = |\mathcal{C}_2|$.

Finally, by Rank-Nullity, $\dim \ker(S) = \dim G_{\lambda_i}(T) - \text{rank}(S) = |\mathcal{C}_1|$. But since $\mathcal{C}_1$ is linearly independent in $\ker(S)$, it follows $\mathcal{C}_1$ must be a basis for $\ker(S)$. By the above remark, a basis vector is in $\mathcal{C}_1$ iff it is in the first r rows of the dot diagram, and thus we are done. $\square$

Theorem 27.6

Let r_j be the number of dots in the jth row of the dot diagram of $T|_{G_{\lambda_i}(T)}$. Then

(i) $r_1 = \dim(V) - \text{rank}(T - \lambda_i \mathbf{1}_V)$

(ii) $r_j = \text{rank}(T - \lambda_i \mathbf{1}_V)^{j-1} - \text{rank}(T - \lambda_i \mathbf{1}_V)^j, \quad j > 1$

Proof. By the previous theorem we have

$$r_1 + \cdots + r_j = \dim \ker(T - \lambda_i \mathbf{1}_V)^j = \dim(V) - \text{rank}(T - \lambda_i \mathbf{1}_V)^j$$

Thus $r_1 = \dim(V) - \text{rank}(T - \lambda_i \mathbf{1}_V)$, and for $j > 1$ we have

$$r_j = (r_1 + \cdots + r_j) - (r_1 + \cdots + r_{j-1})$$

$$= [\dim(V) - \operatorname{rank}(T - \lambda_i \mathbf{1}_V)^j] - [\dim(V) - \operatorname{rank}(T - \lambda_i \mathbf{1}_V)^{j-1}]$$
$$= \operatorname{rank}(T - \lambda_i \mathbf{1}_V)^{j-1} - \operatorname{rank}(T - \lambda_i \mathbf{1}_V)^j$$

□

Corollary 27.7

The dot diagram of $T|_{G_{\lambda_i}(T)}$ is uniquely determined.

Remark 27.8. Friedberg has loads of examples of using these results to compute the Jordan canonical form, but I will omit them here.

Finally, since the sizes of the Jordan blocks are uniquely determined, we can connect Jordan form back to the minimal polynomial:

Theorem 27.9

Let $T : V \to V$ be a linear operator. Suppose the characteristic polynomial of T splits and $\lambda_1, \ldots, \lambda_k$ are the distinct eigenvalues of T. Consider a Jordan canonical form of T, and let p_i be the size of the largest Jordan block corresponding to λ_i. Then the minimal polynomial of T is given by

$$p_T(t) = (t - \lambda_1)^{p_1} \cdots (t - \lambda_k)^{p_k}$$

Proof. Consider each generalized eigenspace $G_{\lambda_i}(T)$, which is T-invariant. Since each Jordan block corresponds exactly to a cycle of generalized eigenvectors, it means the longest cycle has length p_i. But this means there exists v such that $(T - \lambda_i I)^{p_i - 1}(v)$ is an eigenvector, so $(T - \lambda_i I)^{p_i - 1} \neq 0$. On the other hand we know $(T - \lambda_i I)^{p_i} = 0$ on $G_{\lambda_i}(T)$. Note that the characteristic polynomial of $T|_{G_{\lambda_i}(T)}$ is $(t - \lambda_i)^{\dim G_{\lambda_i}(T)}$. Hence the minimal polynomial of $T|_{G_{\lambda_i}(T)}$ must divide this, and must be $(t - \lambda_i)^{p_i}$.

Then putting this together we must have $(t - \lambda_1)^{p_1} \ldots (t - \lambda_k)^{p_k} \mid p_T(t)$. But the generalized eigenspaces give us a direct sum decomposition so for any $v \in V$ we can write $v = v_1 + \cdots + v_k$ where $v_i \in G_{\lambda_i}(T)$. But since $(T - \lambda_i I)^{p_i}(v_i) = 0$ for each i, it follows $(T - \lambda_1 I)^{p_1} \ldots (T - \lambda_k I)^{p_k} = 0$, and hence the minimal polynomial is $(t - \lambda_1)^{p_1} \ldots (t - \lambda_k)^{p_k}$. □

Lecture 28 June 1

Review

Lecture 29 June 3

Review

Lecture 30 June 5

No class

Appendix A An Aside on Diagram Chasing and Exact Sequences

The professor seems pretty fond of using diagram chasing to represent the maps and spaces we are talking about so I thought it might be good to include a section on it. This is a thing you see a lot in algebra (particularly homological algebra and category theory), where you can write a lot of the relations more compactly in terms of commutative diagrams or sequences. This allows you to read off properties much more easily (but you should ask an algebraist to tell you more about that).

Let's first practice reading these diagrams:

Example 30.1

Let V be a vector space over a field F, and let W be a subspace.

$$W \xhookrightarrow{i} V \xrightarrow{\pi} V/W$$

Here the first arrow from W to V represents the natural inclusion map i, defined by $i(w) = w$ for all $w \in W$. The second arrow represents the canonical quotient map π, defined by $\pi(v) = v + W$ for all $v \in V$. Here I specified the maps, but sometimes they may be omitted if it is obvious, or if there is a natural to consider. For example, we will almost always only consider the quotient map when we see an arrow from V to V/W.

Example 30.2

Of course, there is a special subspace we can consider, namely the kernel. Recall the first isomorphism theorem says that given a linear map $T : V \to W$, we have $V/\ker T \simeq \operatorname{im} T$. Then the above diagram becomes

$$\ker T \xhookrightarrow{i} V \xrightarrow{\pi} V/\ker T \simeq \operatorname{im} T$$

But we can chain together a bunch of maps and spaces without it having any meaning. But the diagrams we had in class take a very special form.

Definition 30.3 (Exact and Short Exact Sequences) — An **exact sequence** is a sequence of vector spaces

$$V_0 \xrightarrow{f_1} V_1 \xrightarrow{f_2} V_2 \xrightarrow{f_3} \dots \xrightarrow{f_n} V_n$$

where each arrow is a linear map and satisfies $\mathrm{im}(f_i) = \ker(f_{i+1})$.

A **short exact sequence** is a diagram of the following form:

$$0 \longrightarrow A \xrightarrow{S} B \xrightarrow{T} C \longrightarrow 0$$

where 0 is the zero vector space, and sequence is exact.

Remark 30.4. Note that it is implied that the arrows to and from the 0 vector spaces are the 0 maps. From 0 to A, we must map 0 to 0 since the map is linear, but that is the only element in the zero vector space. From C to 0, the only possibly linear map is the zero map that takes everything to 0.

Proposition 30.5

$0 \xrightarrow{\alpha} A \xrightarrow{S} B \xrightarrow{T} C \xrightarrow{\beta} 0$ is a short exact sequence iff S is injective, T is surjective, and $\mathrm{im}(S) = \ker(T)$

Proof. ($\implies$) By the definition of exact sequences, we must have $\mathrm{im}(\alpha) = \ker(S)$. (Here α is the zero map). But $\mathrm{im}(\alpha) = \{0\}$, so S is injective. Similarly we must have $\mathrm{im}(T) = \ker(\beta)$. But β maps everything in C to 0 so $\ker(\beta) = C$, and T is surjective.

($\impliedby$) Again we have $\mathrm{im}(\alpha) = \{0\}$ and $\ker(S) = \{0\}$ by injectivity, as well as $\ker(\beta) = C$ and $\mathrm{im}(T) = C$ by surjectivity. By assumption $\mathrm{im}(S) = \ker(T)$. Thus we have a short exact sequence. □

Example 30.6

Now if we look at the example from before, we see it is a short exact sequence:

$$0 \longrightarrow W \xhookrightarrow{i} V \xrightarrow{\pi} V/W \longrightarrow 0$$

The inclusion map i is clearly injective, and the quotient map π is also surjective (for every $v + W \in V/W$ we can pick out some v that maps to it). And we have proved before that $\ker(\pi) = W$ (or convince yourself) and clearly $\mathrm{im}(i) = W$. Thus the sequence is short exact.

Why is this useful? Short exact sequences of vector spaces have a special property (called being split) which makes it so that it essentially encodes the fact that $V \simeq W \oplus V/W$, which I do not want to get into too much detail. Essentially it comes from what I mentioned before that vector spaces have enough structure that allows us to define really nice concepts like bases and dimension, making it so that any two vector spaces with the same dimension are isomorphic.

Notice this also implies for the second example $V \simeq \ker(T) \oplus \mathrm{im}(T)$, but they are NOT equal.

Definition 30.7 (Dual Sequence) — Given a short exact sequence

$$0 \longrightarrow A \xrightarrow{S} B \xrightarrow{T} C \longrightarrow 0$$

the dual sequence is given by

$$0 \longleftarrow A^* \xleftarrow{S^*} B^* \xleftarrow{T^*} C^* \longleftarrow 0$$

We take the dual of each vector space (note the only linear functional on the zero vector space is the zero functional, so it is essentially still the zero vector space), and since the dual map flips directions, so do the arrows. The dual sequence is important because of the following property:

Theorem 30.8

Suppose all the vector spaces are finite dimensional. The dual sequence of a short exact sequence is also short exact. That is, if

$$0 \longrightarrow A \xrightarrow{S} B \xrightarrow{T} C \longrightarrow 0$$

is a short exact sequence, then

$$0 \longrightarrow C^* \xrightarrow{T^*} B^* \xrightarrow{S^*} A^* \longrightarrow 0$$

is also a short exact sequence.

The proof of this theorem is not too difficult, but a little longer than desirable, and it is more enlightening to see some of the interesting facts we can deduce using the theorem.

Example 30.9

The theorem tells us the dual sequence

$$0 \longrightarrow (V/W)^* \xrightarrow{\pi^*} V^* \xrightarrow{i^*} W^* \longrightarrow 0$$

is also exact. In particular this means π^* is injective and i^* is surjective, which makes it pretty easy to guess what the maps are. As an exercise, you can check i^* is the restriction map from linear functionals defined on V to the functionals defined on W, and π^* injects $(V/W)^*$ to the annihilator $W^0 \subseteq V^*$.

The above theorem also gives an easy proof of the earlier theorem we had in class:

Theorem 30.10

Suppose V and W are finite dimensional spaces over F and $T : V \to W$ is linear. Then

1. T is injective $\iff$ T^* is surjective.
2. T is surjective $\iff$ T^* is injective.

Proof. 1. ($\implies$) Assume T is injective.

Claim: The sequence

$$0 \longrightarrow V \xrightarrow{T} W \xrightarrow{\pi} W/\operatorname{im}(T) \longrightarrow 0$$

is exact.

Pf: We have by assumption T is injective, and the quotient map π is also surjective. Now we have $\ker(\pi) = \operatorname{im}(T)$ so the sequence is exact.

Then the dual sequence

$$0 \longrightarrow (W/\operatorname{im}(T))^* \xrightarrow{\pi^*} W^* \xrightarrow{T^*} V^* \xrightarrow{\beta} 0$$

is exact. By the definition of an exact sequence we must have $\operatorname{im}(\pi^*) = \ker(T^*)$. But since the dual sequence is exact π^* is injective, meaning $\operatorname{im}(\pi^*) \simeq (W/\operatorname{im}(T))^*$. Thus $(W/\operatorname{im}(T))^* \simeq \ker(T^*)$, as the professor claimed. But also by exactness we have $\operatorname{im}(T^*) = \ker(\beta) = V^*$. Thus T^* is surjective.

($\impliedby$) Assume T^* is surjective. This direction requires you to be comfortable with the properties of the annihilator. (See Axler Linear Algebra Done Right.)

Claim: The sequence

$$0 \longrightarrow (W/\operatorname{im}(T))^* \xrightarrow{\pi^*} W^* \xrightarrow{T^*} V^* \longrightarrow 0$$

is exact.

Pf: $(W/\operatorname{im}(T))^*$ injects into $\operatorname{im}(T)^0 \subseteq W^*$ via π^* (exercise). T^* is by assumption surjective. We have $\ker(T^*) = (\operatorname{im}(T))^0$. Thus the sequence is exact.

Then its dual sequence

$$0 \longrightarrow V^{**} \xrightarrow{T^{**}} W^{**} \xrightarrow{\pi^{**}} (W/\operatorname{im}(T))^{**} \longrightarrow 0$$

is exact. But recall for finite-dimensional spaces there is a natural isomorphism between a space and its double dual (and hence the maps and their double dual). Thus we can take the isomorphism and get rid of the double stars to get

$$0 \xrightarrow{\alpha} V \xrightarrow{T} W \xrightarrow{\pi} W/\operatorname{im}(T) \longrightarrow 0$$

is exact. Then by exactness $\{0\} = \operatorname{im}(\alpha) = \ker(T)$ so T is injective.

2. Same idea, except now look at the sequence and corresponding dual sequence of

$$0 \longrightarrow \ker(T) \xrightarrow{i} V \xrightarrow{T} W \longrightarrow 0$$

$\square$

Remark 30.11. The quotient space $W/\operatorname{im}(T)$ is also called the **cokernel** of T, denoted $\operatorname{coker}(T)$. This makes the symmetry between the exact sequences used in 1. and 2. more obvious.